

Wrong Models in Ecology II

Patterns in Microbial Communities, Fluctuations and Environmental Stochasticity

Jacopo Grilli, Bengaluru 16/1/2025

Statistical ecology

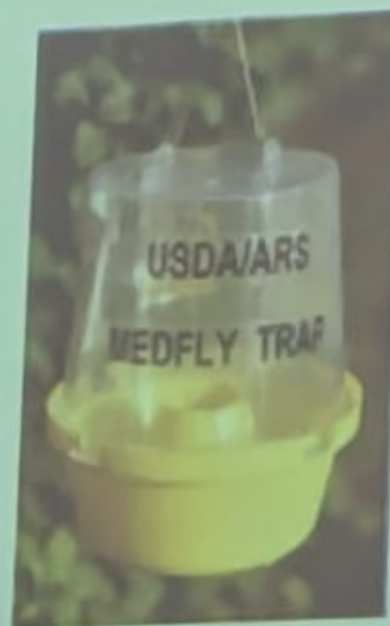
Predict species "fitness" (presence, abundance, reproduction, other measurements...) from environment & own traits

Ecological data

Controlled studies



Natural observation



Data matrix

		Sample					Descriptor			
		a	b	c	d	e	I	II	III	IV
Species	1									
	2									
	3									
	4									
	5									
	6									
	7									
Factor	X									
	Y									
	Z									
	W									

Tends to be noisy and sparse

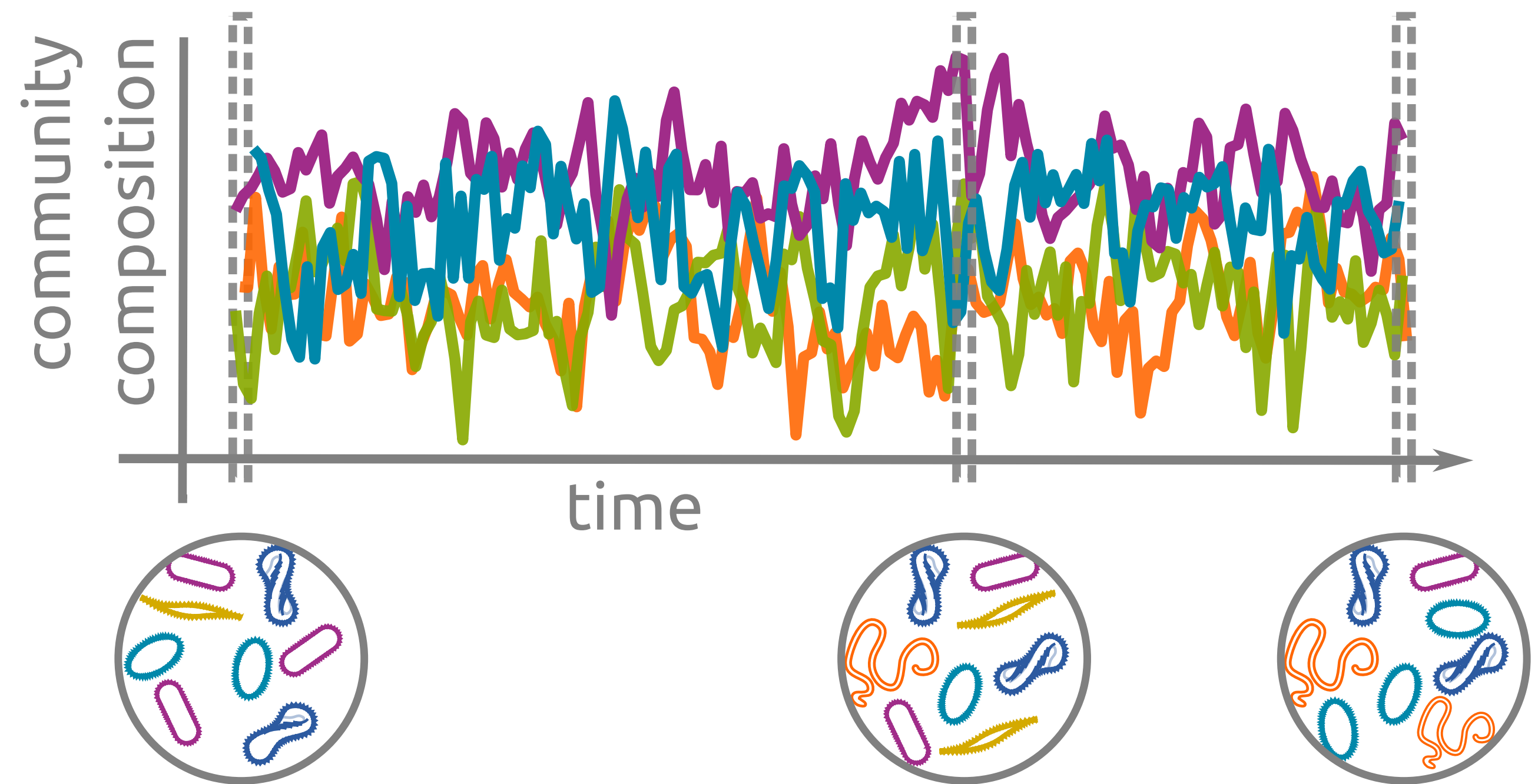
Today

		Sample				
		a	b	c	d	e
Species	1					
	2					
	3					
	4					
	5					
	6					
	7					

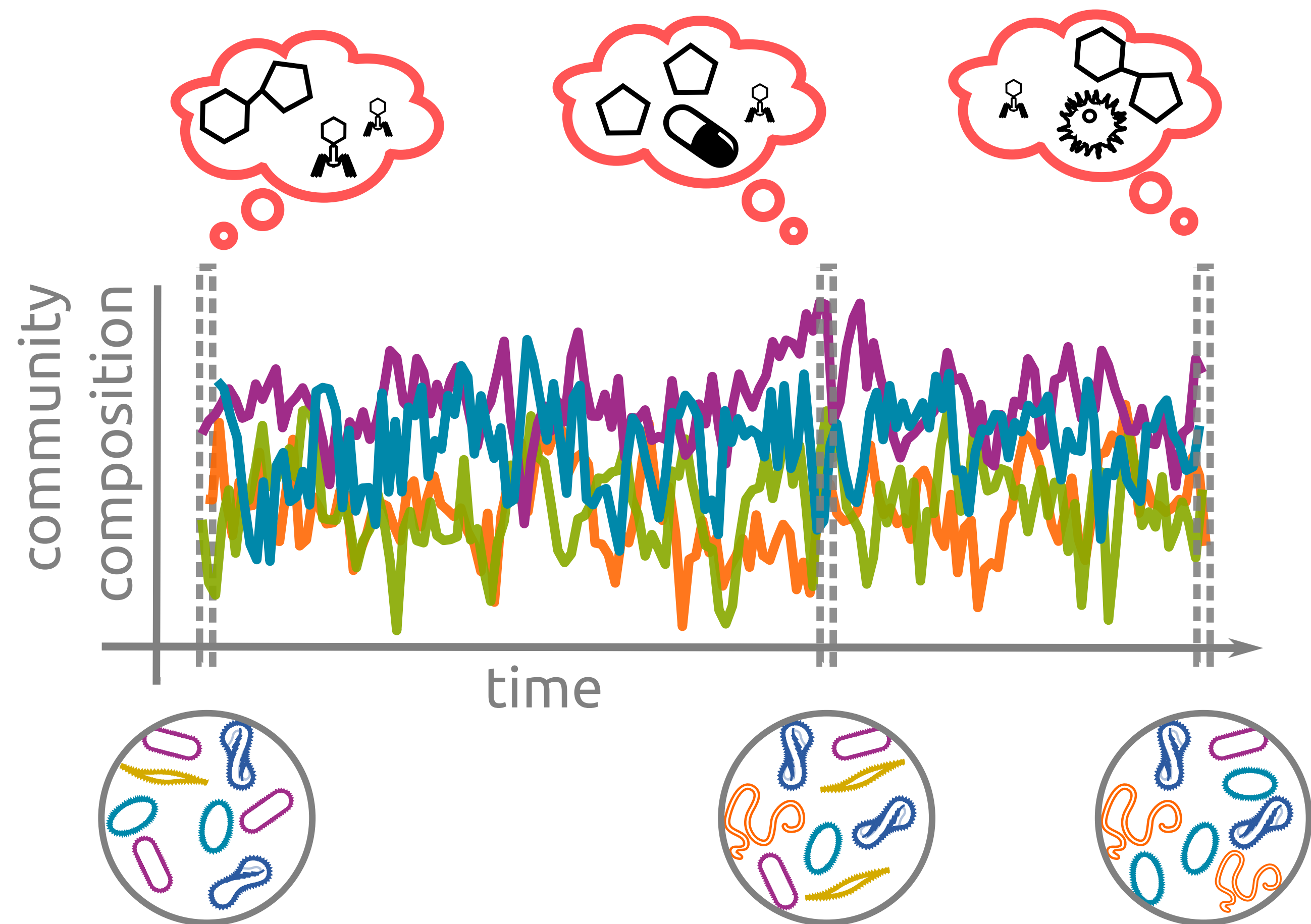
Tends to be noisy and sparse

Data on Microbial Communities

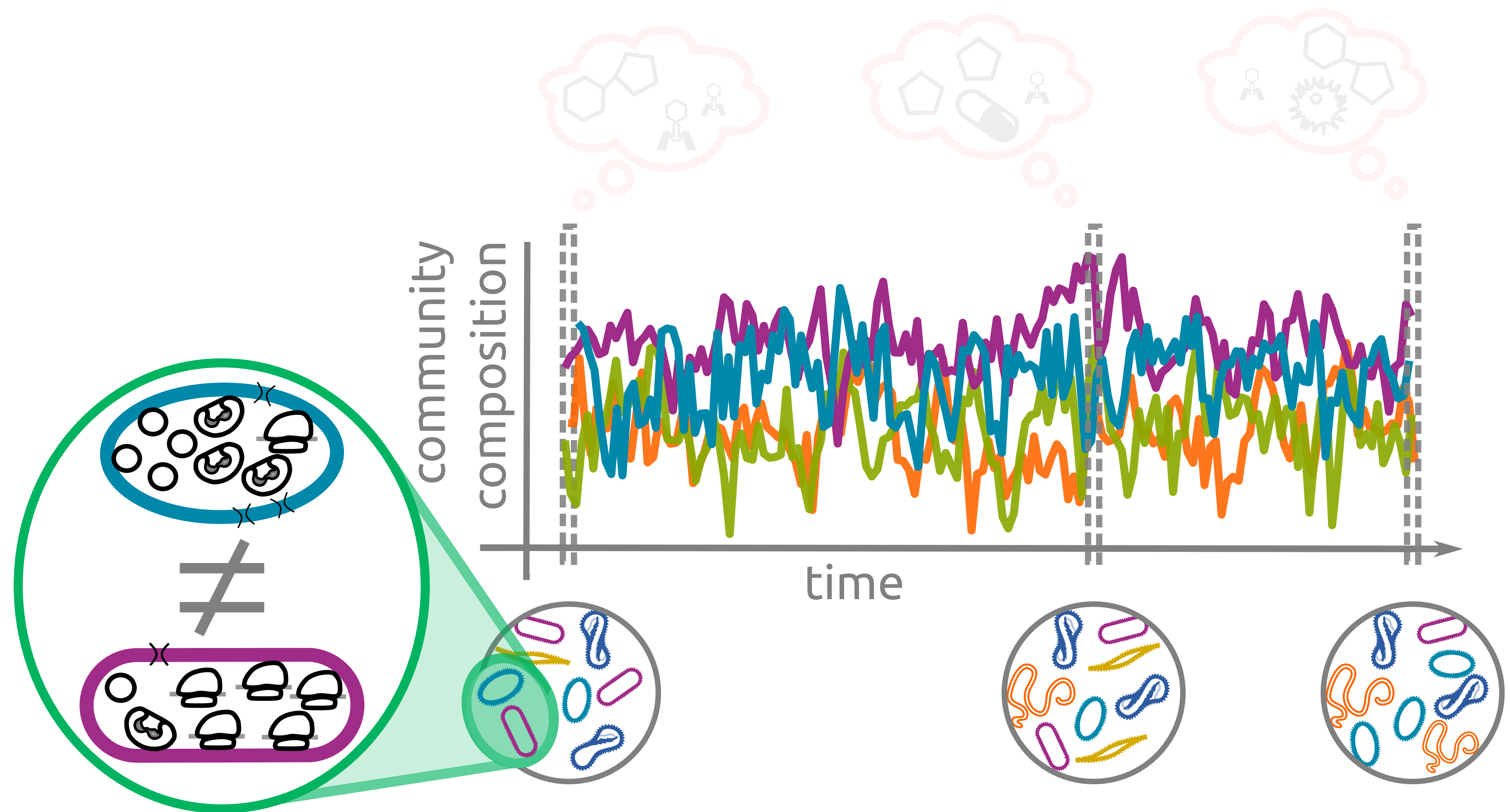
The dynamics of communities is complex



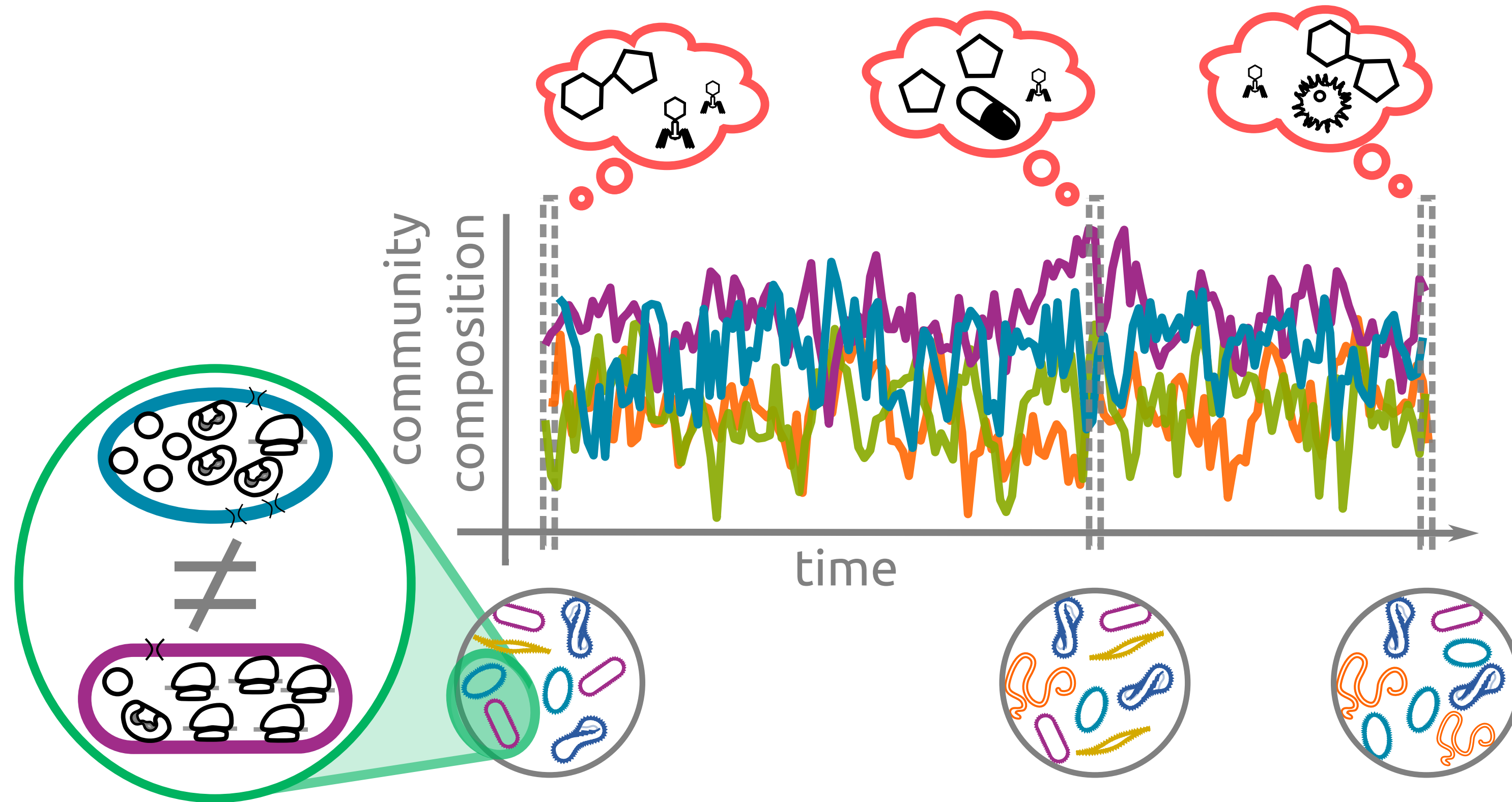
because the environment is high-dimensional



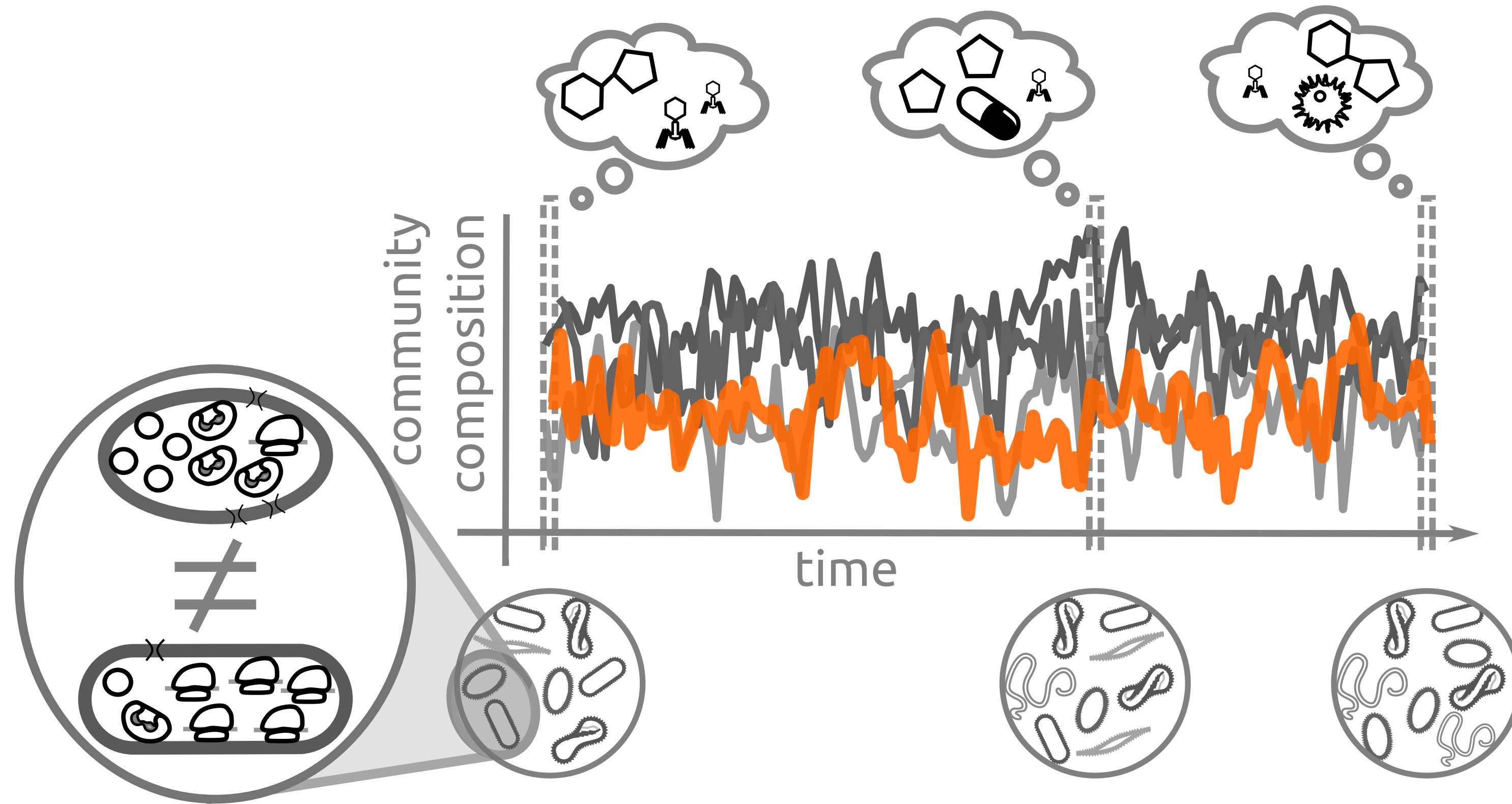
because trait space is high-dimensional



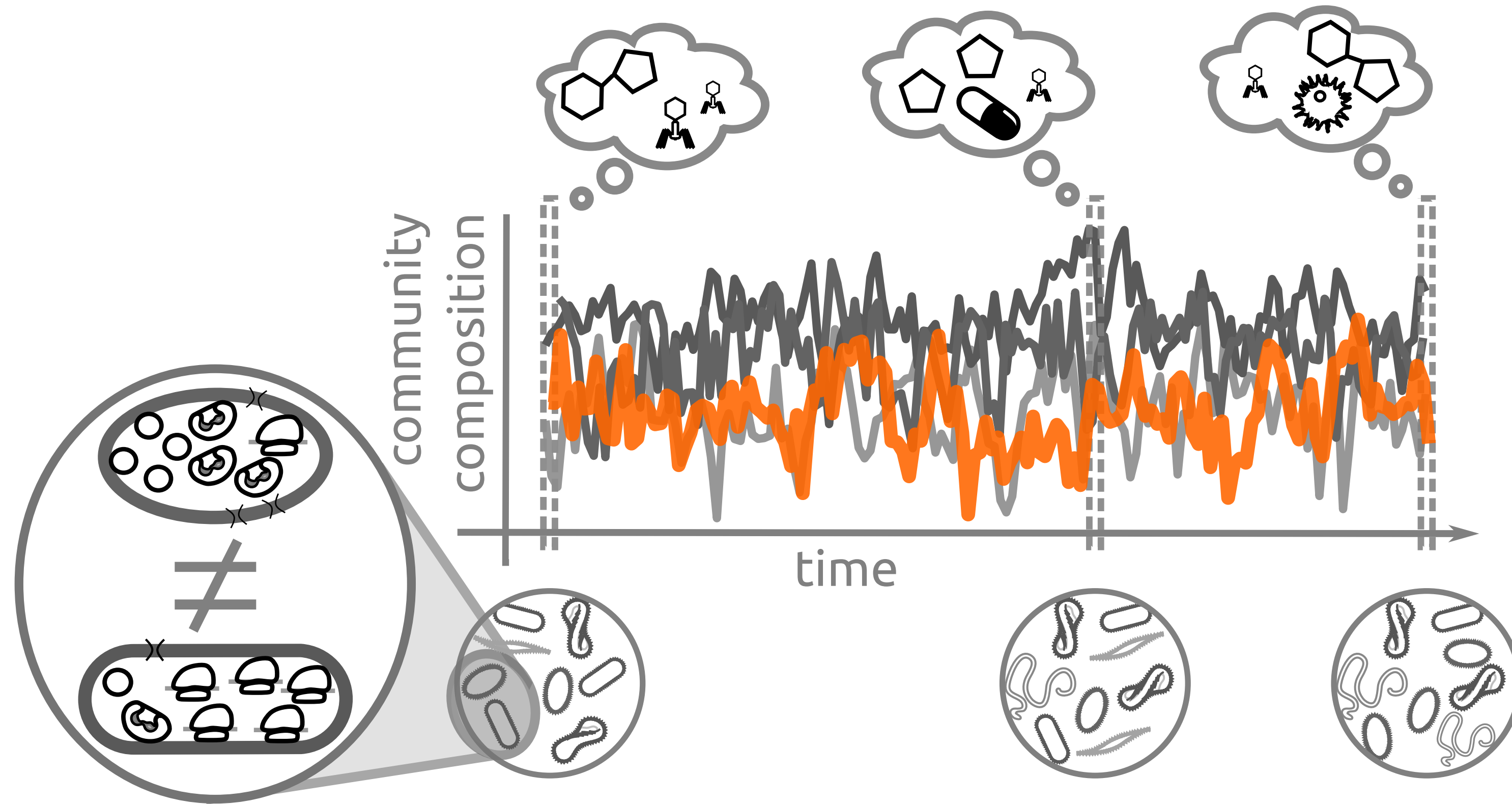
what is the statistical structure of microbial community dynamics?



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what is the statistical structure of microbial community dynamics?



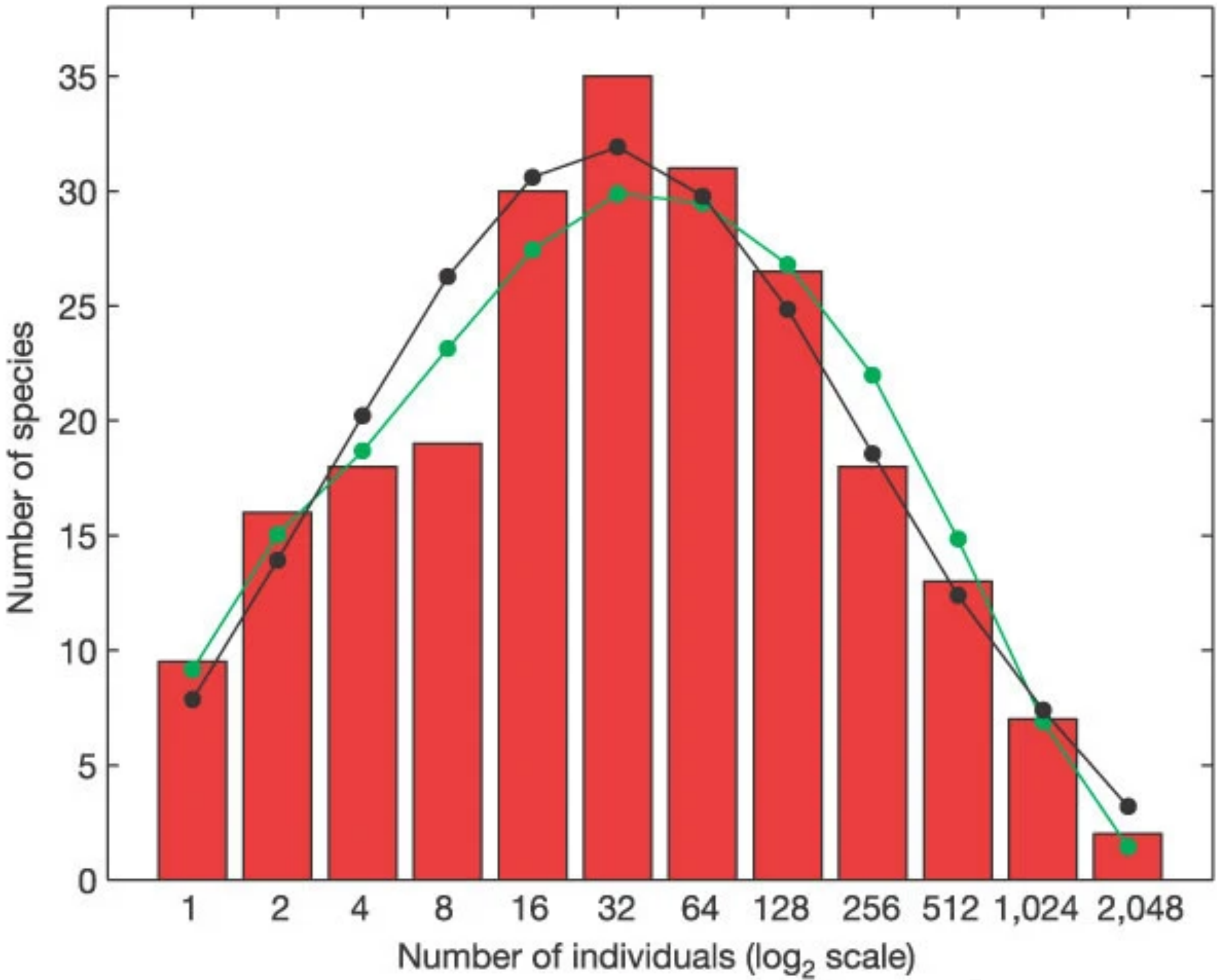
$$P(\text{community}; t) = ?$$

A very basic phenomenology

There are many different “things” (species/
types)

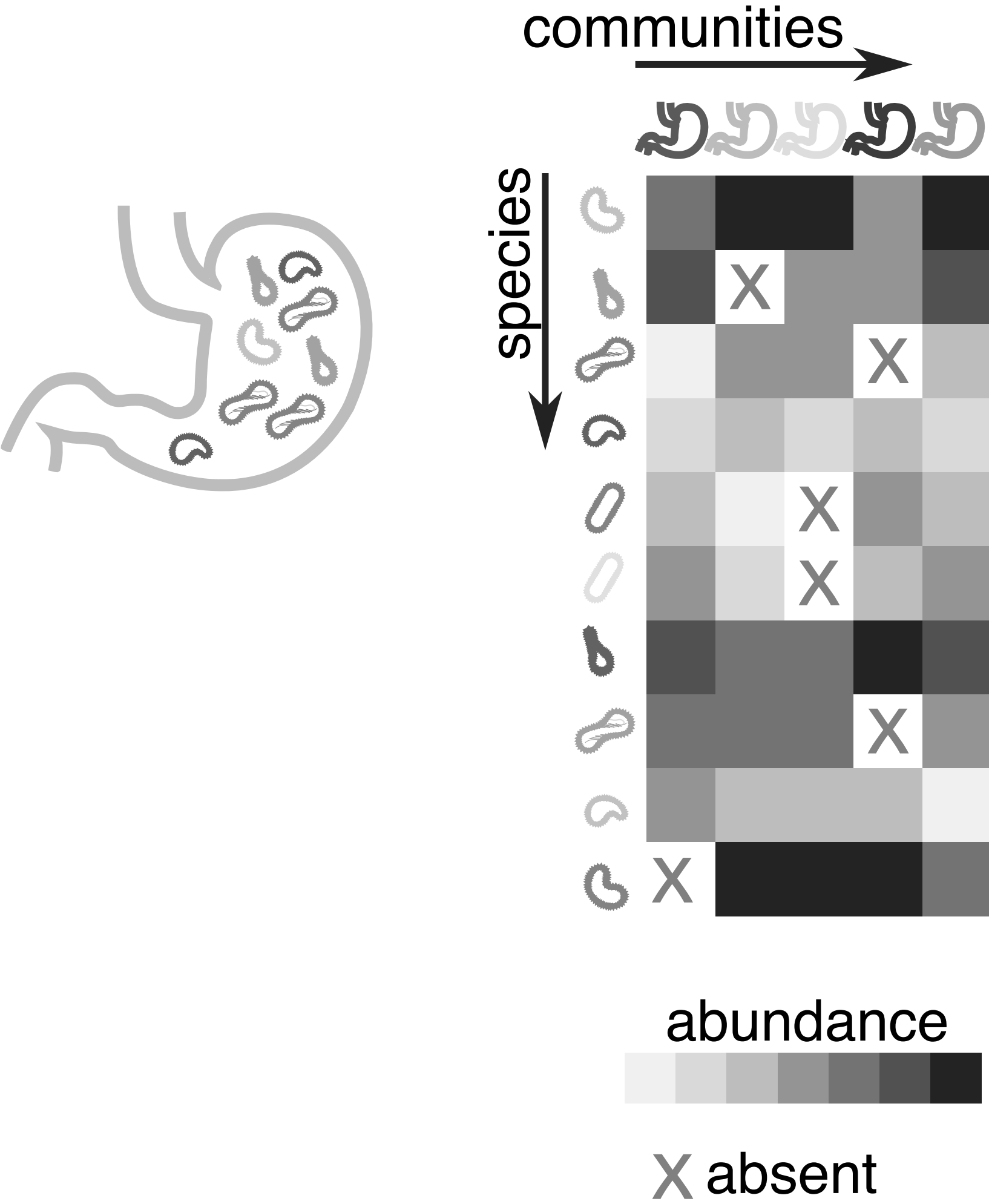
There are “common” and “rare” species

Making the observation more quantitative

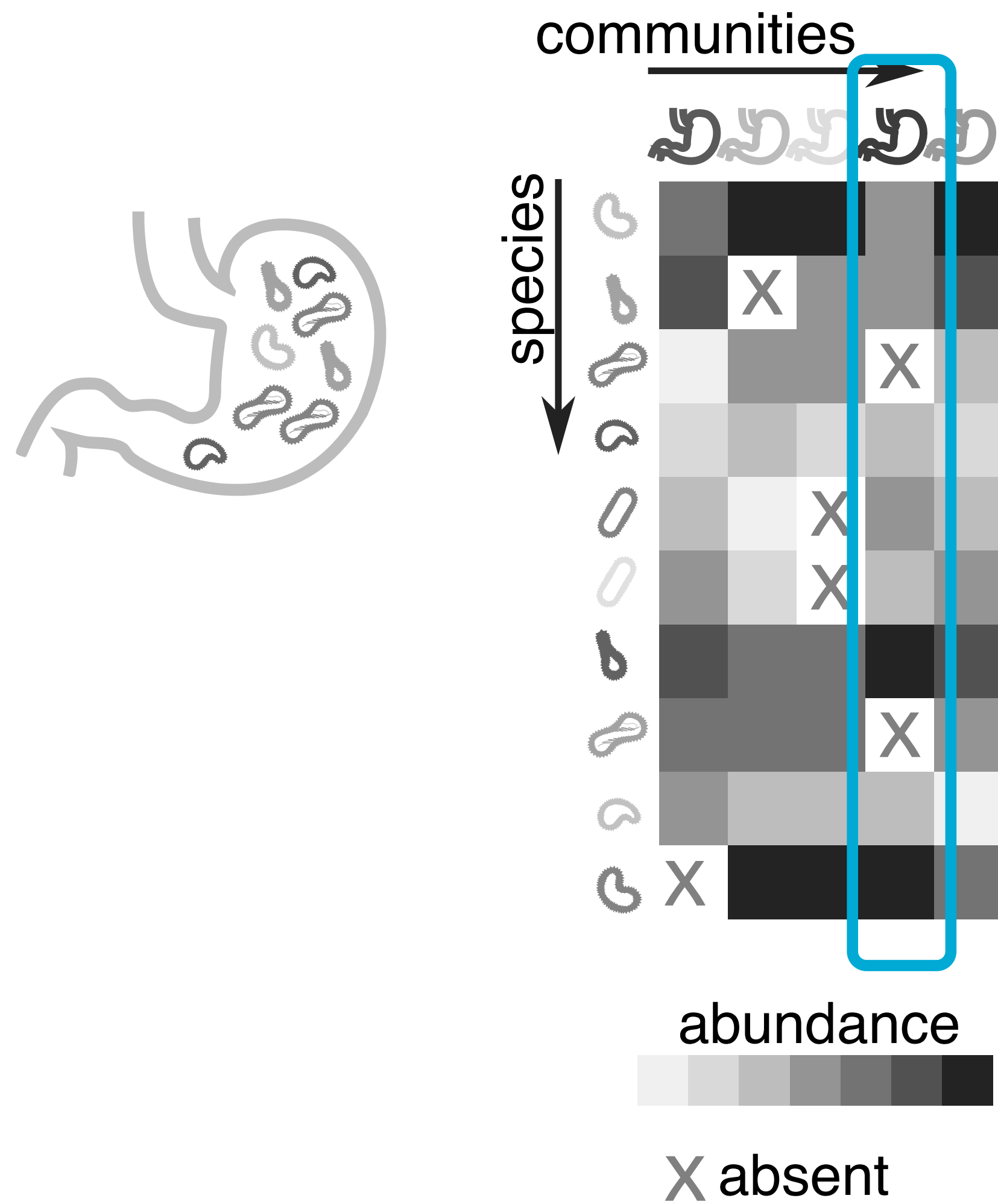


[Volkov et al., 2003]

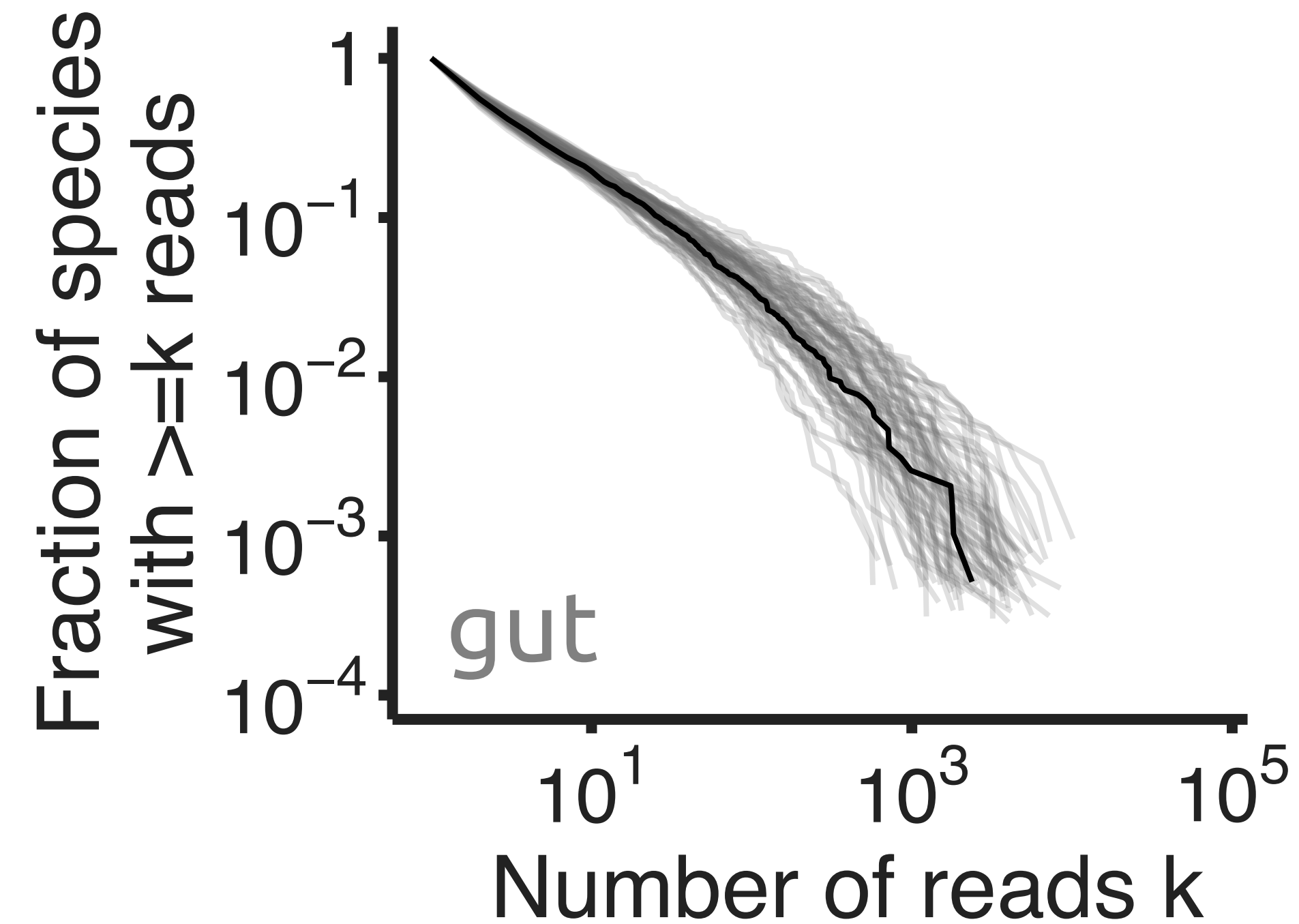
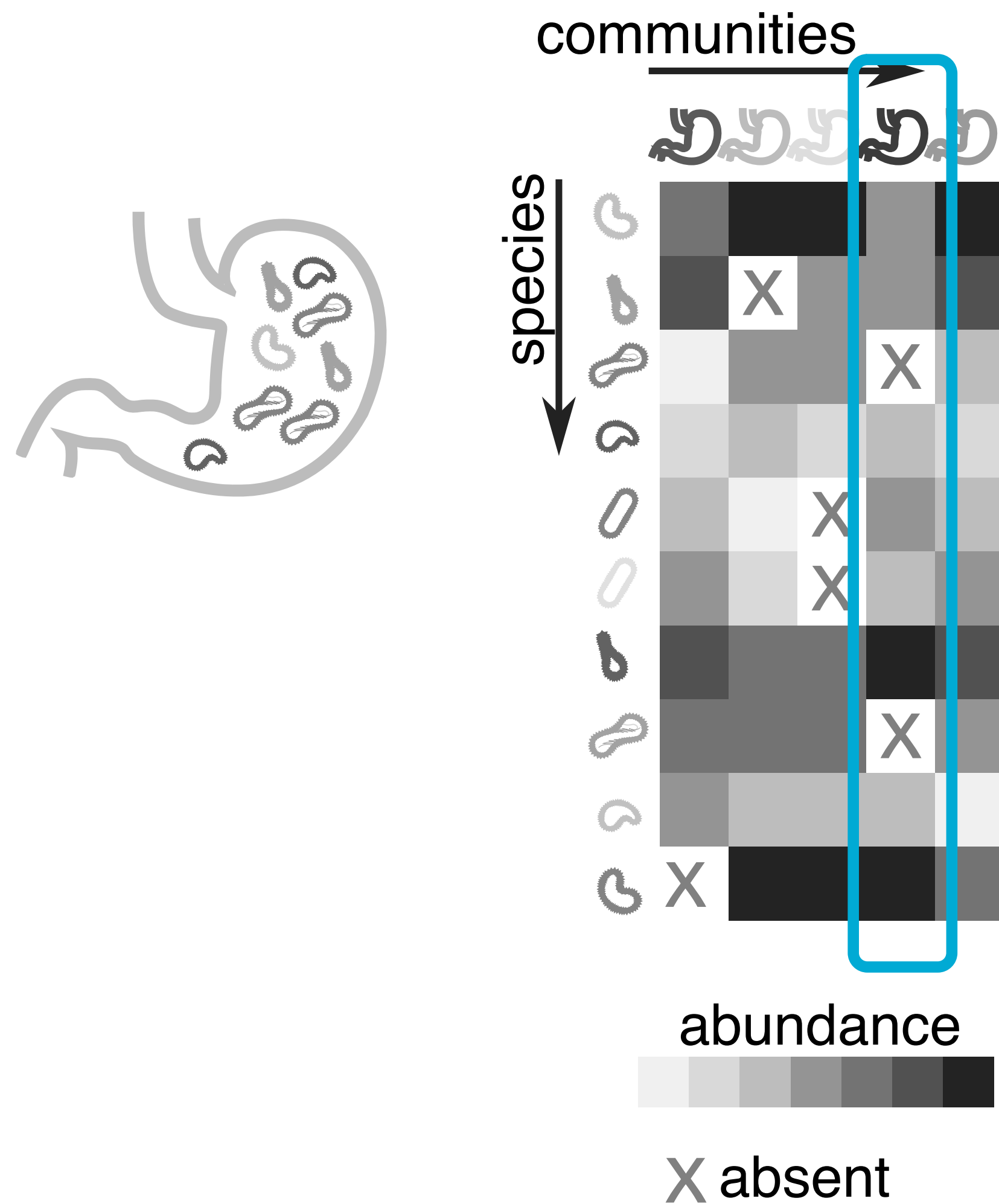
Species Abundance distribution



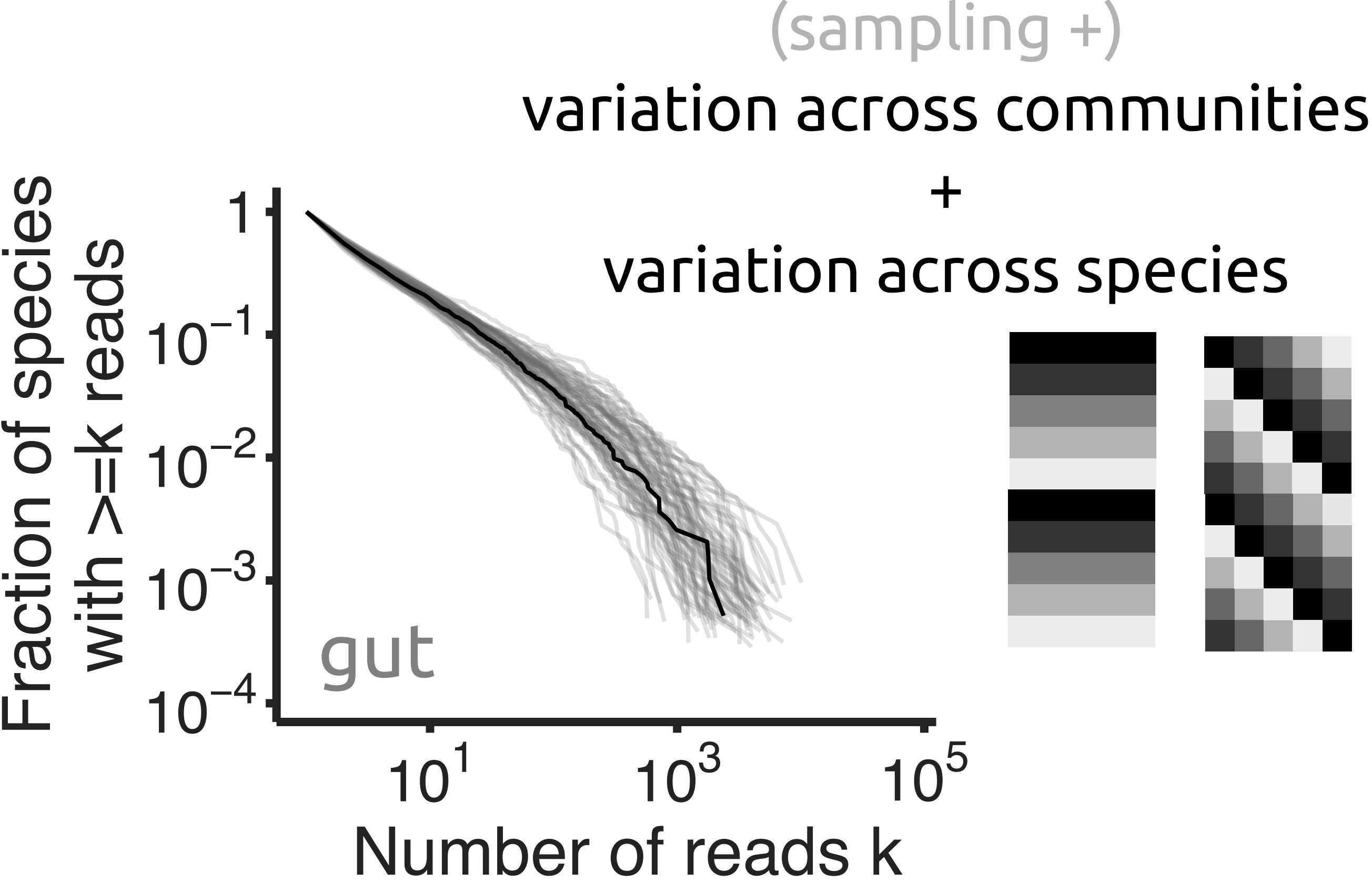
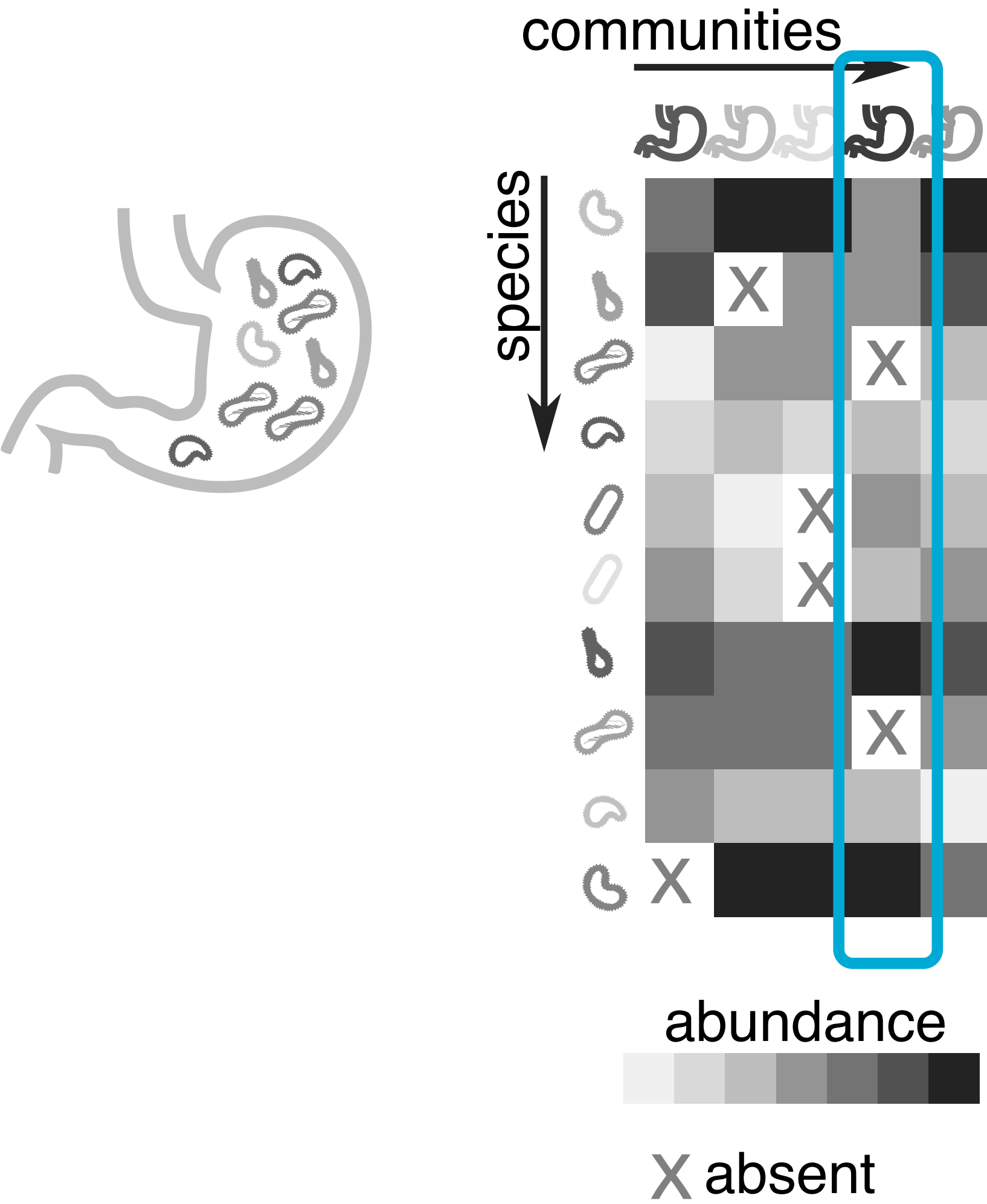
Species Abundance distribution



Species Abundance distribution

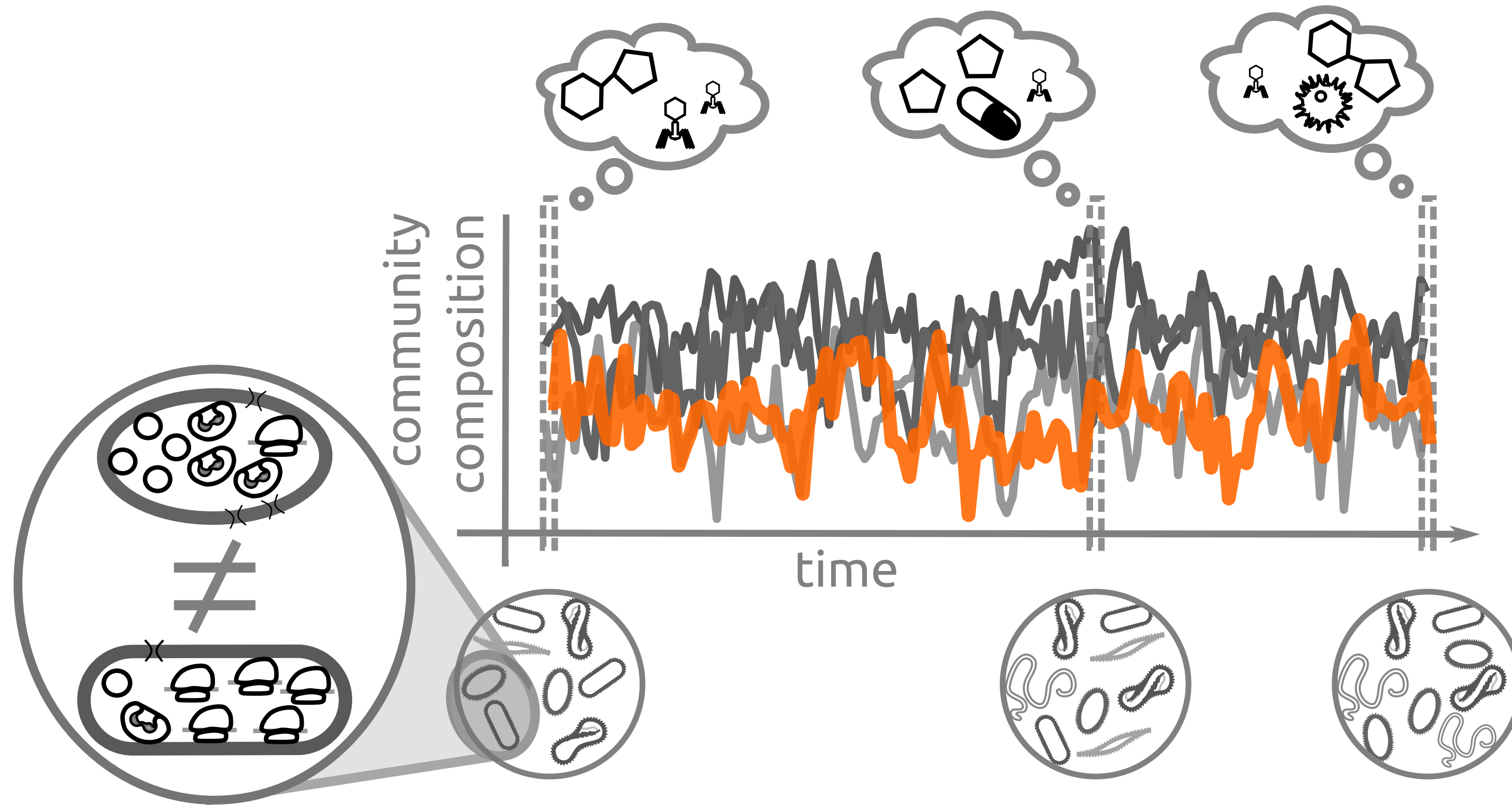


Species Abundance distribution

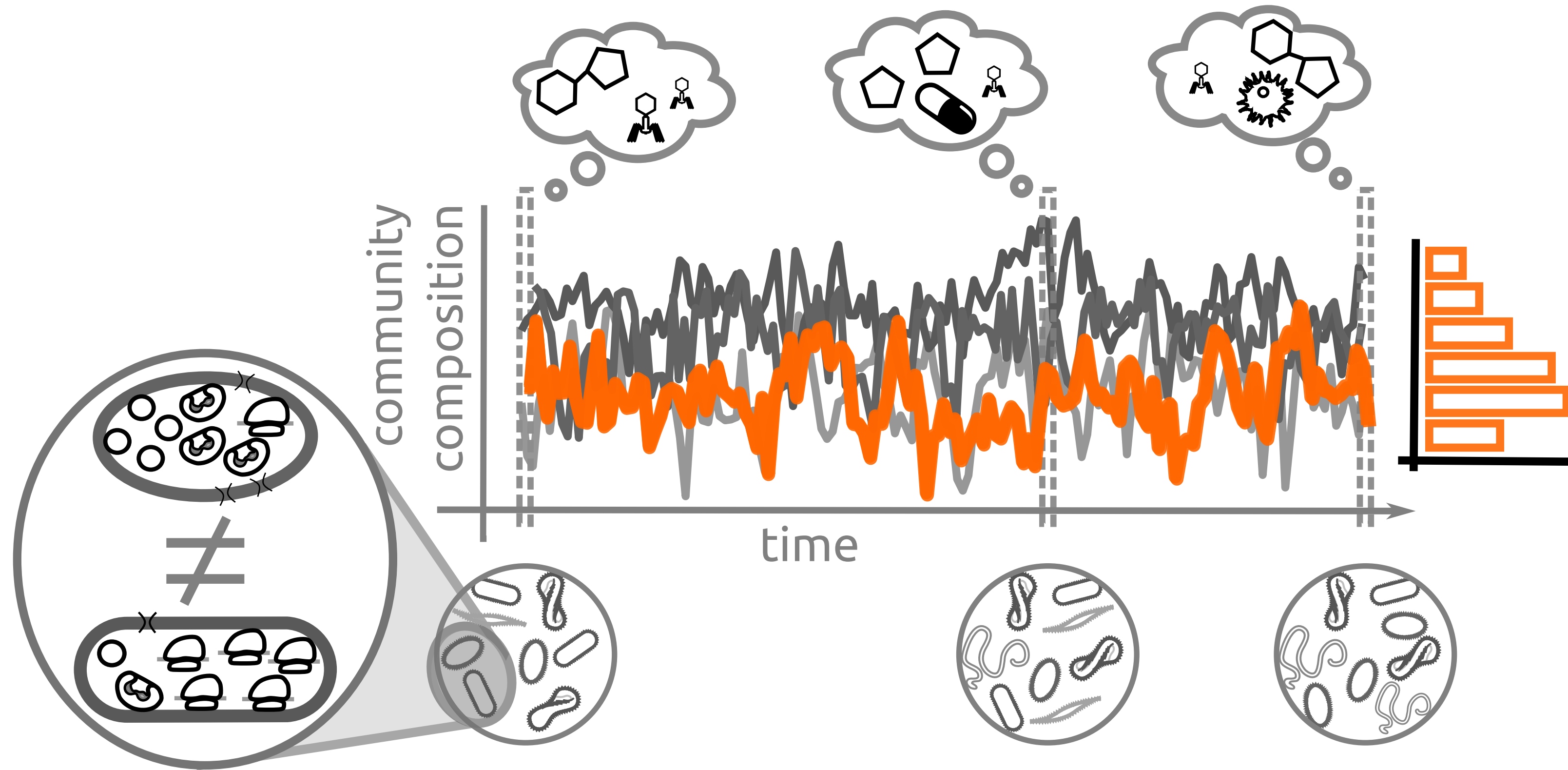


Abundance fluctuations are Gamma distr

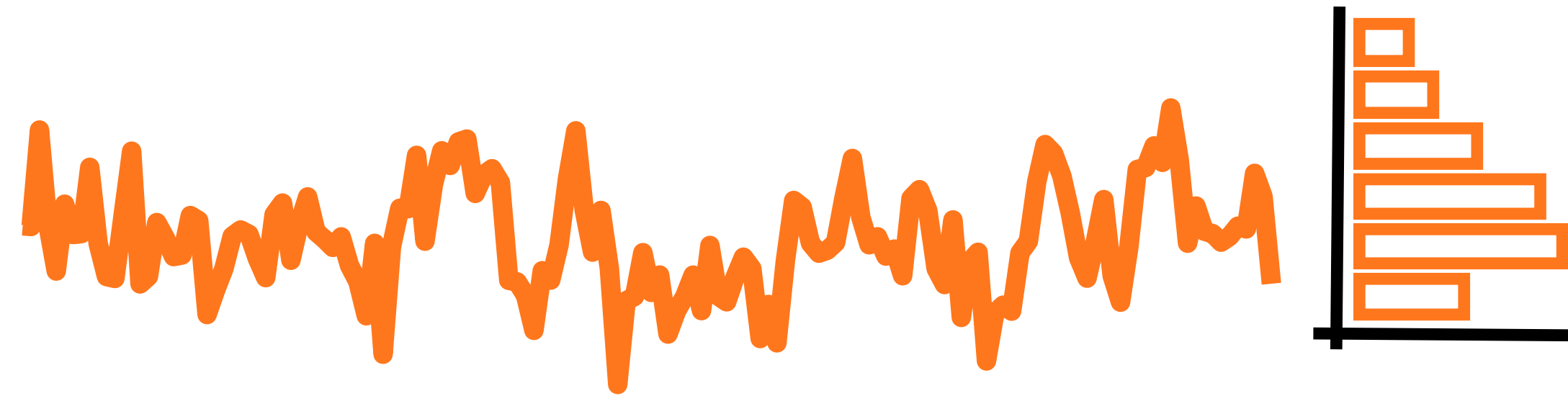
Abundance fluctuations are Gamma distr



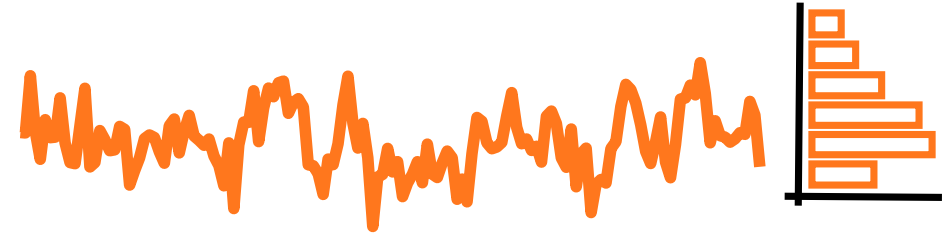
Abundance fluctuations are Gamma distr



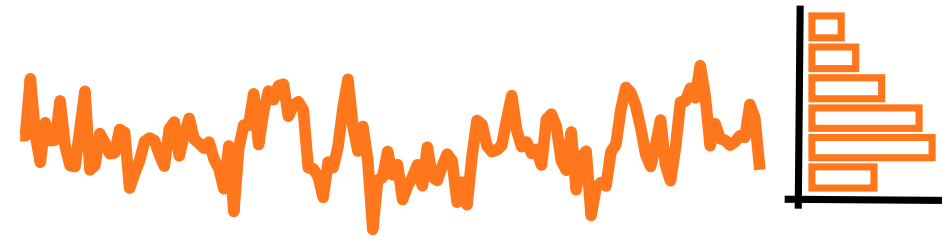
Abundance fluctuations are Gamma distr



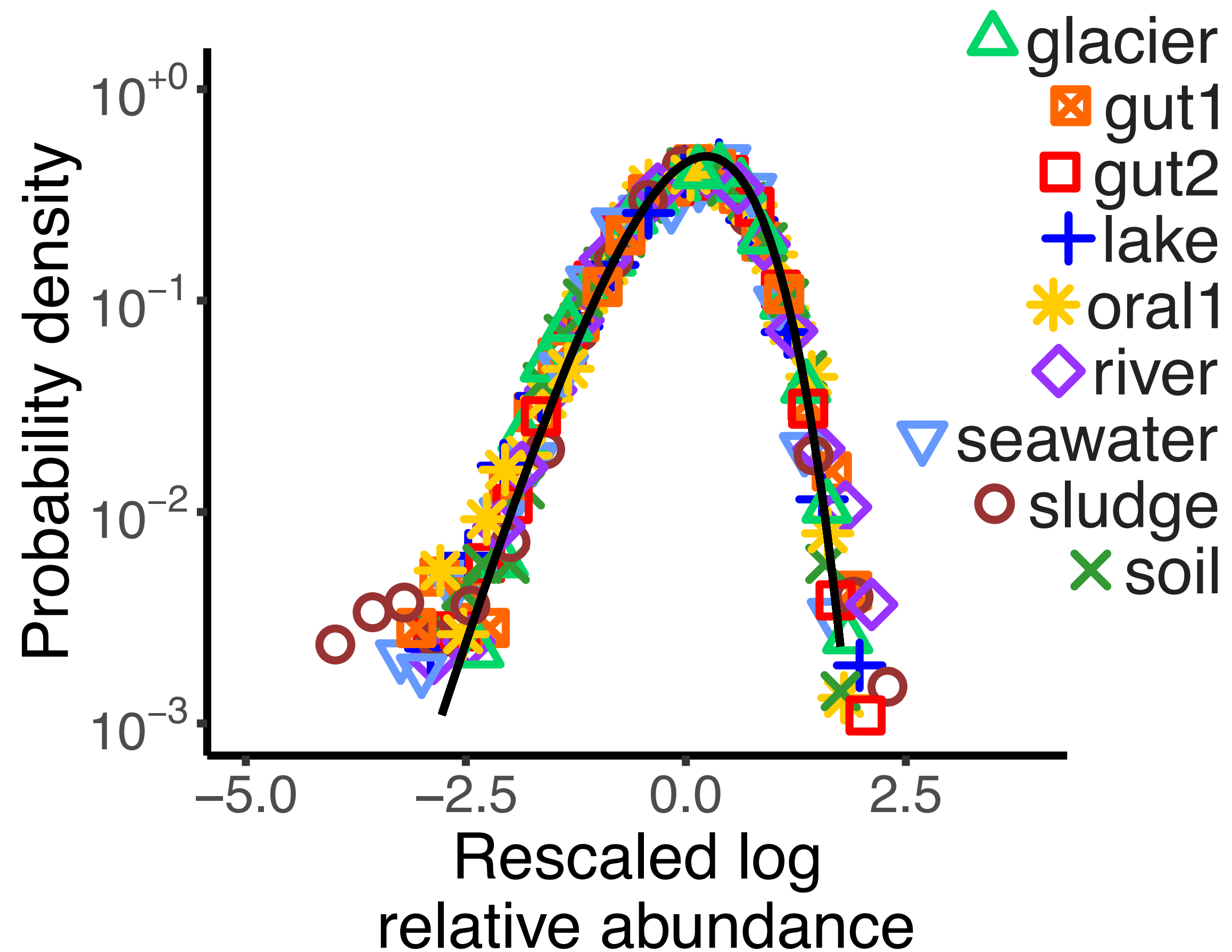
Abundance fluctuations are Gamma distr



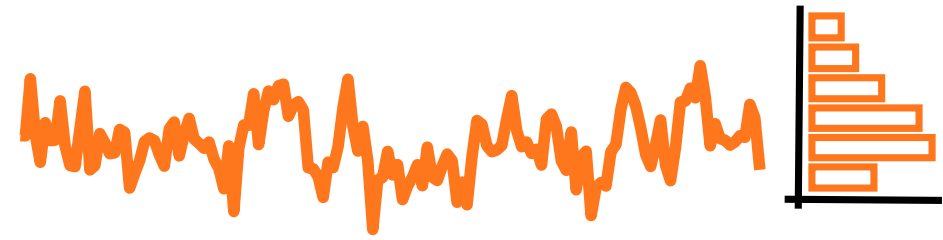
Abundance fluctuations are Gamma distr



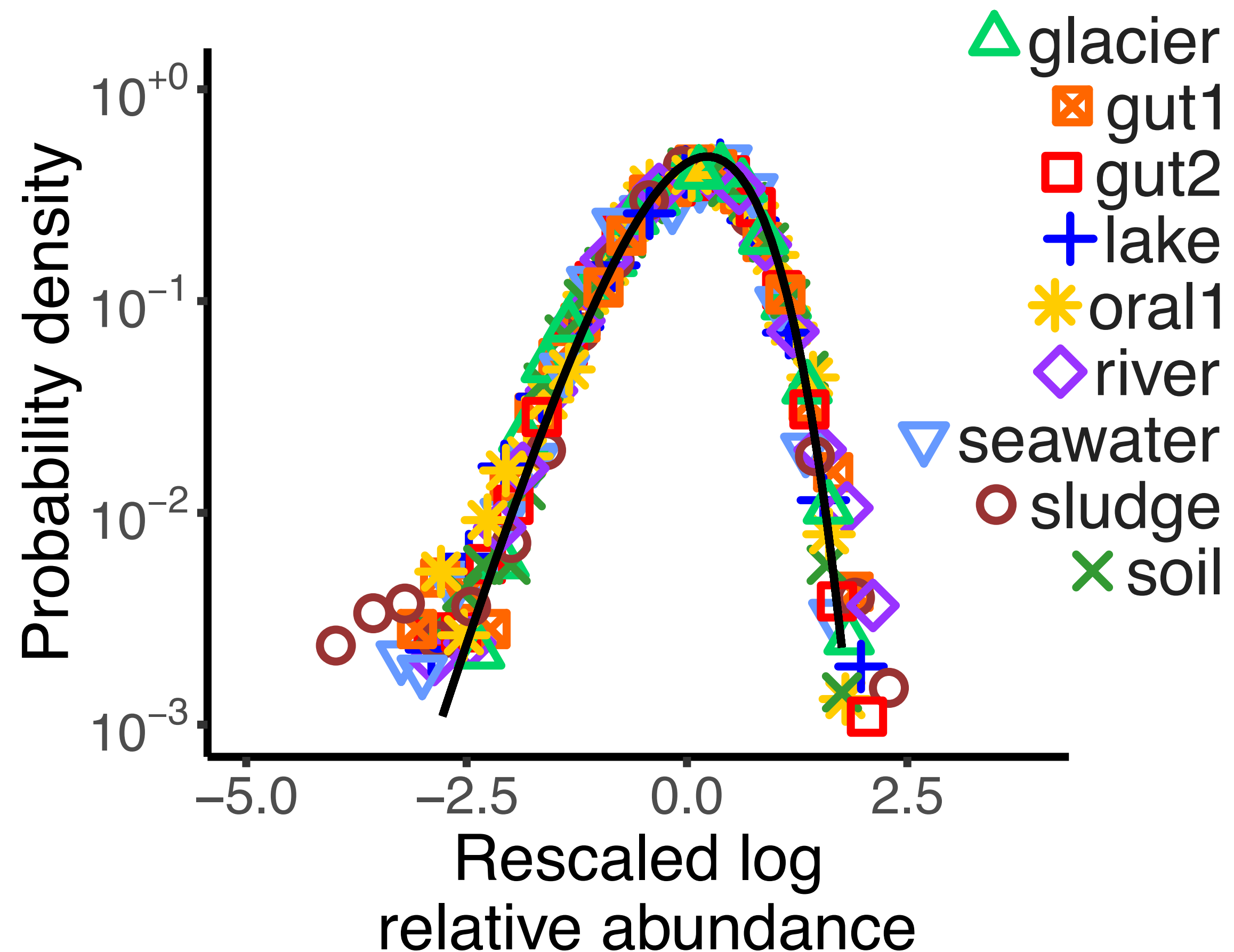
variation across communities
for multiple environments



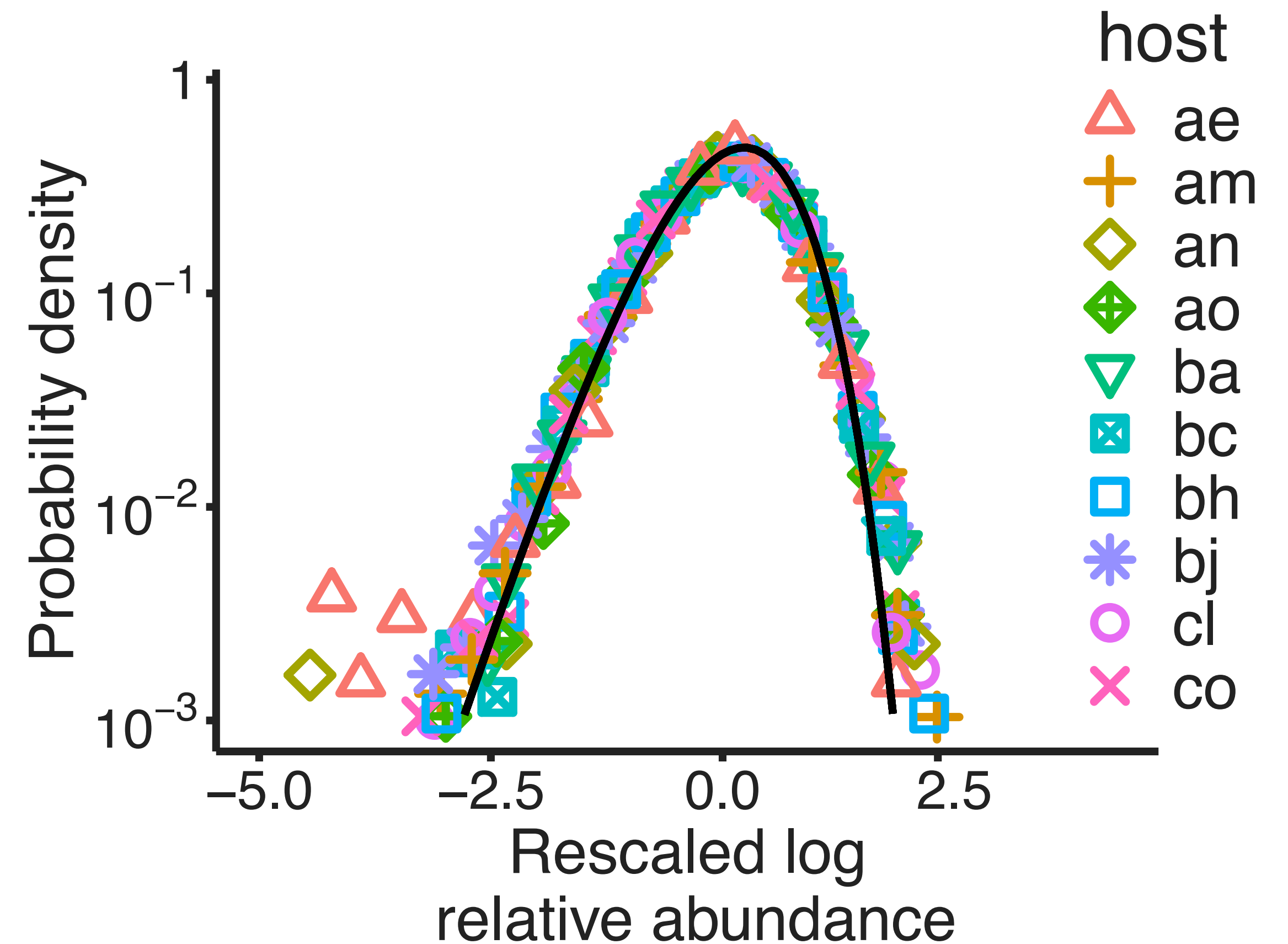
Abundance fluctuations are Gamma distr

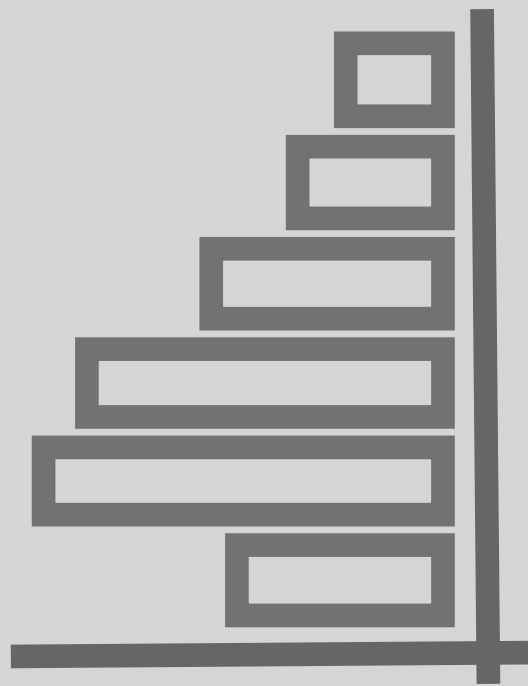
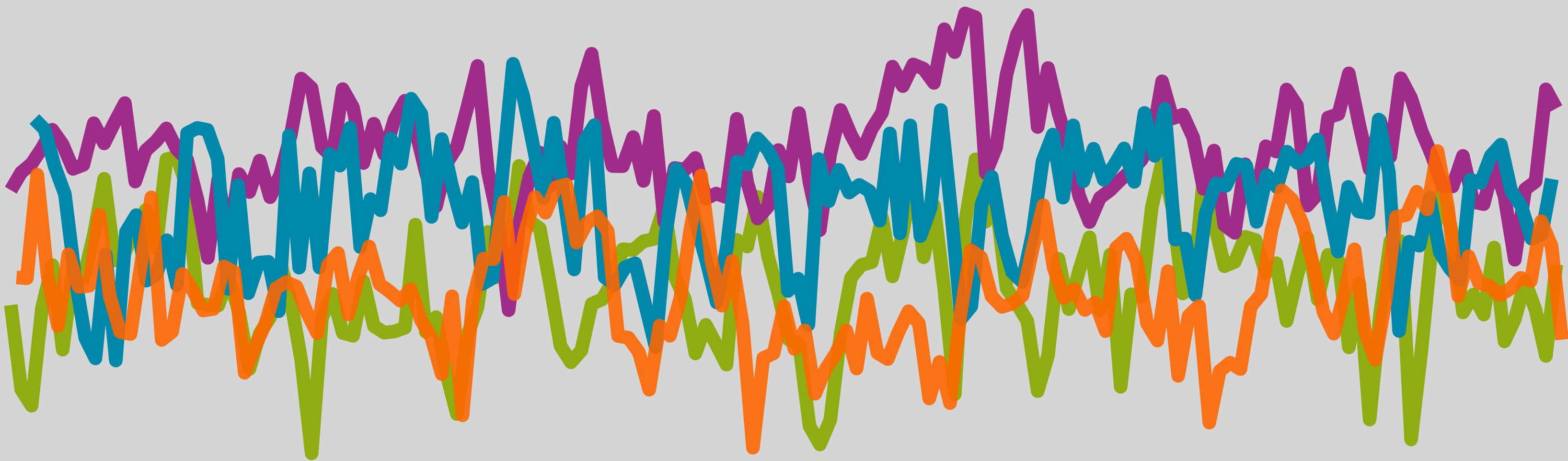


variation across communities
for multiple environments

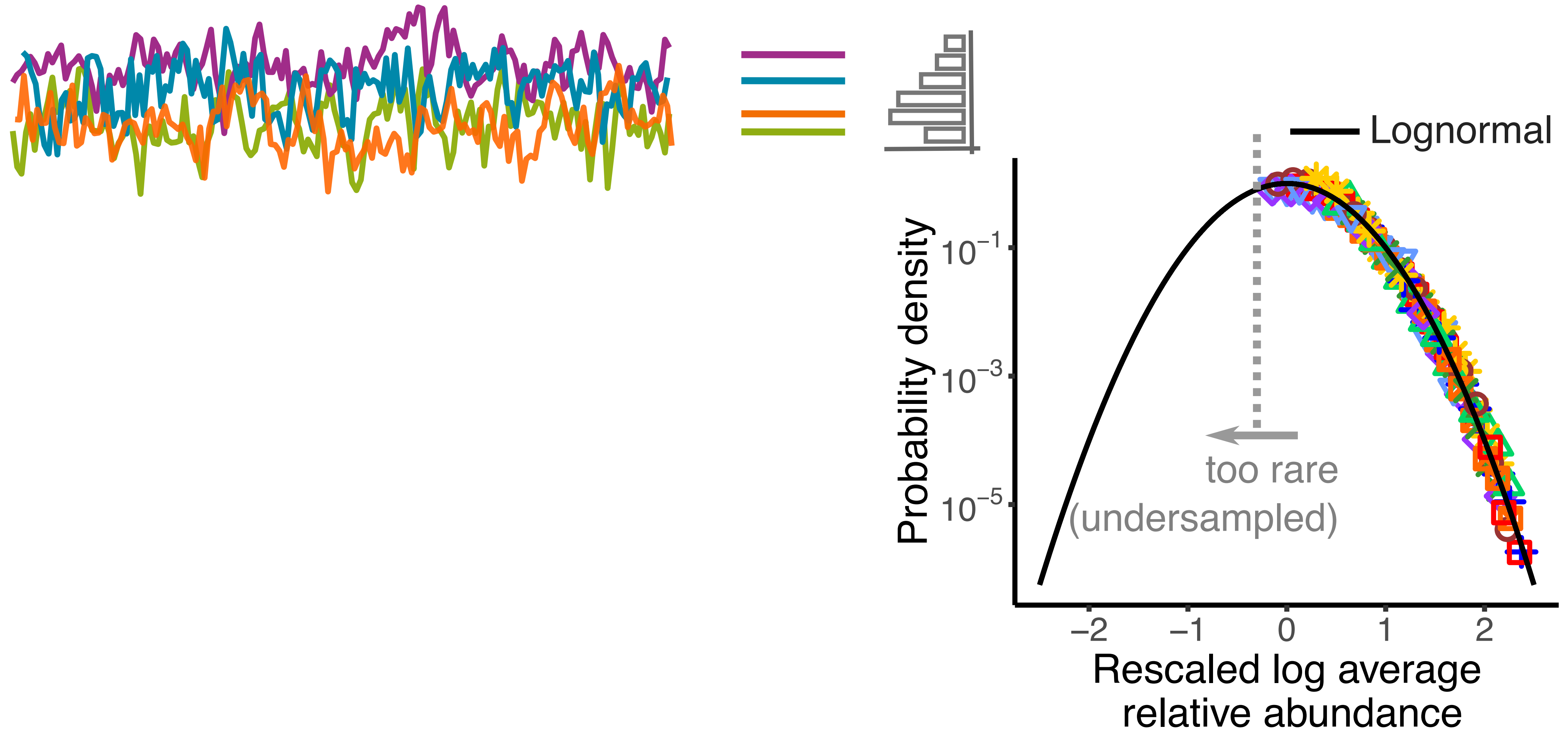


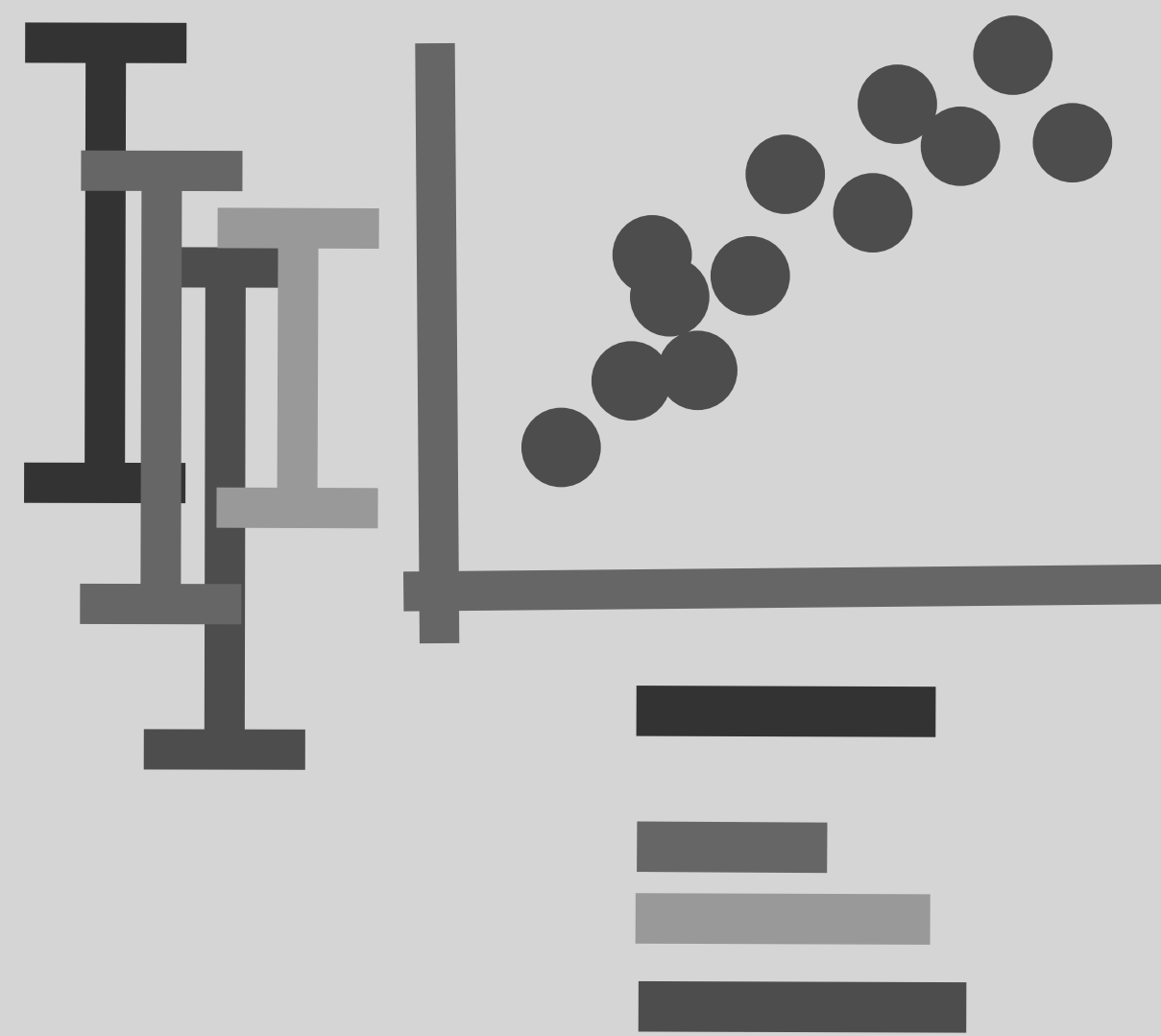
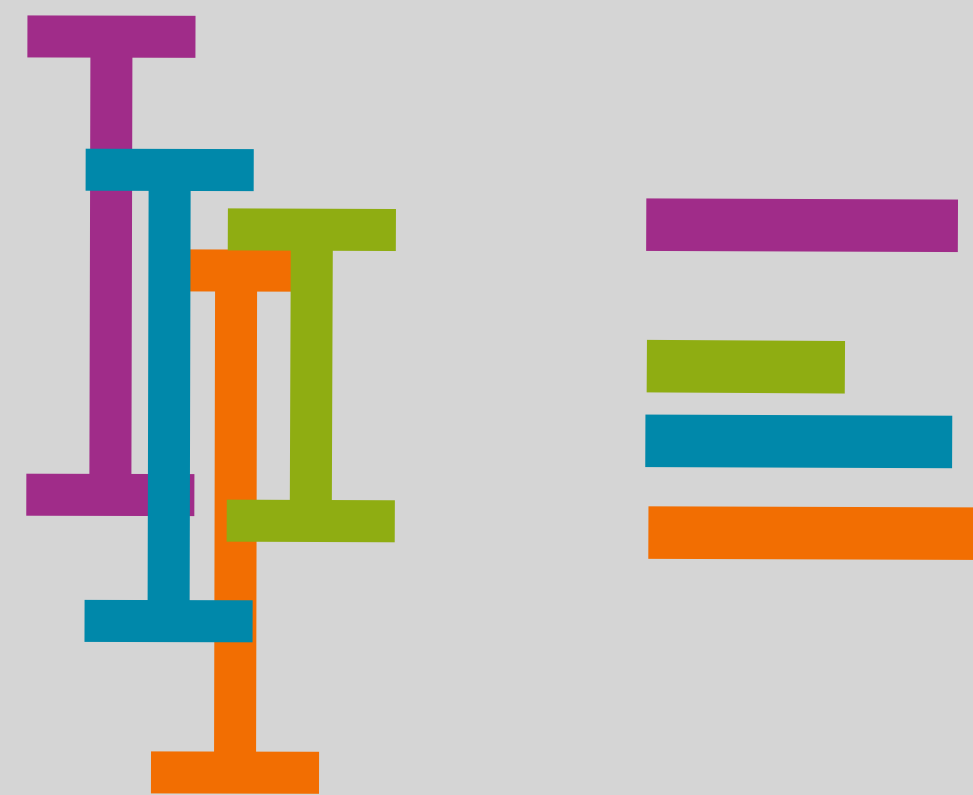
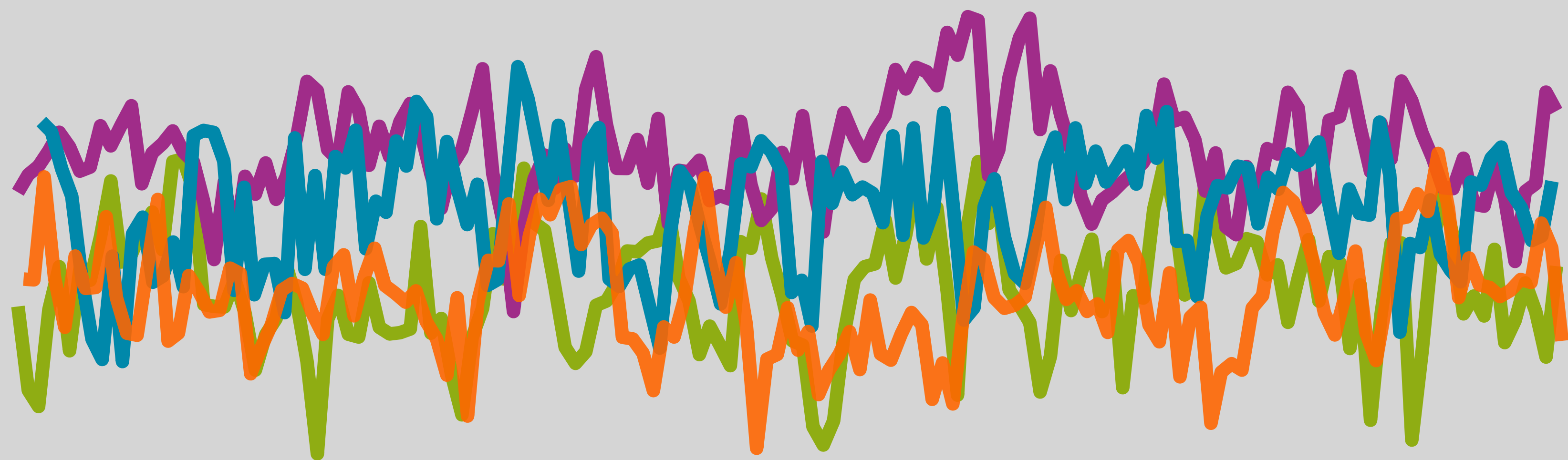
variation over time
(human gut)



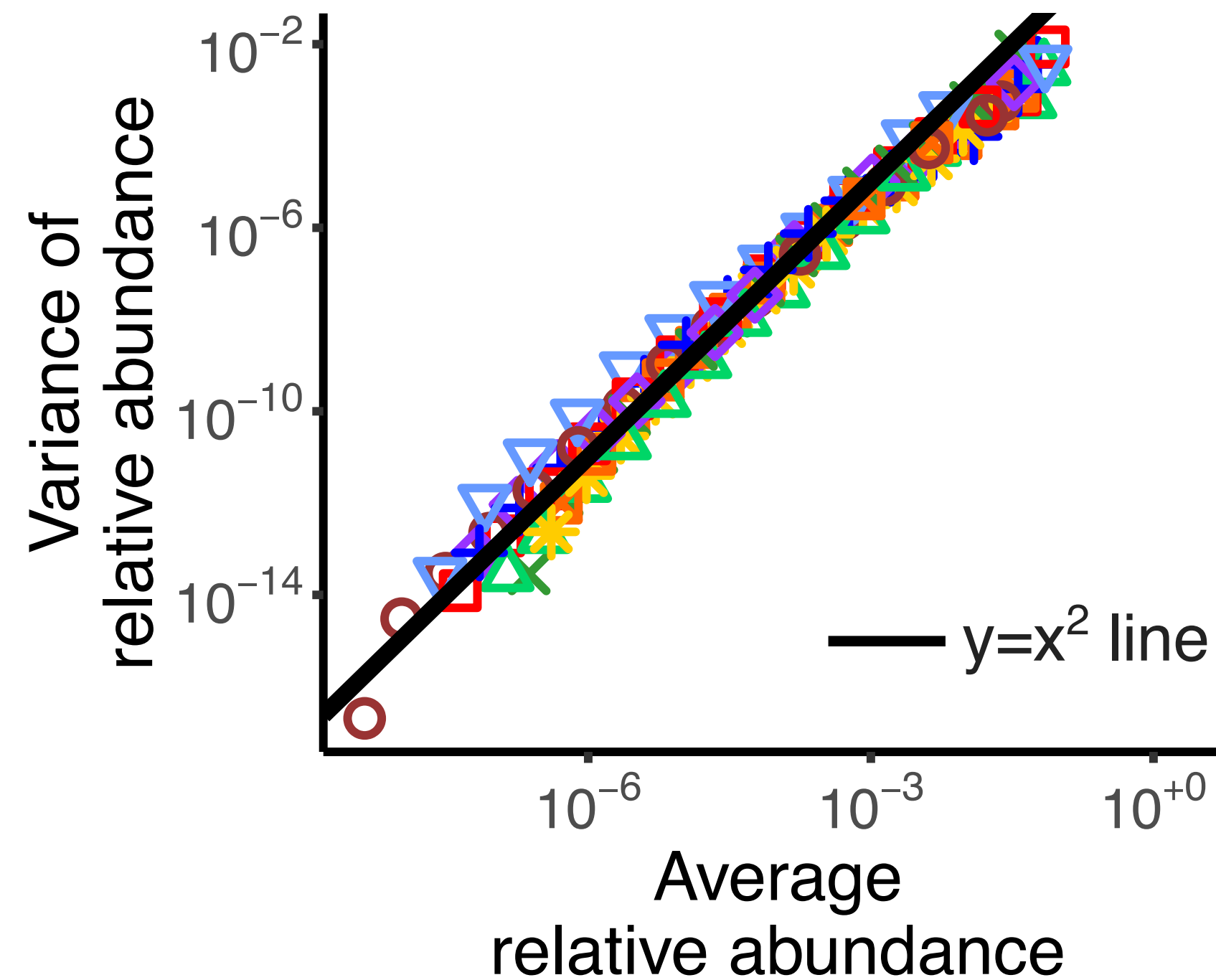
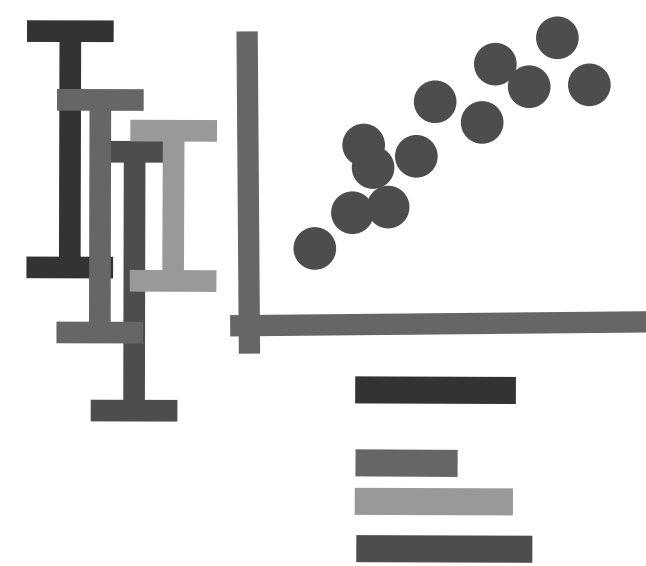
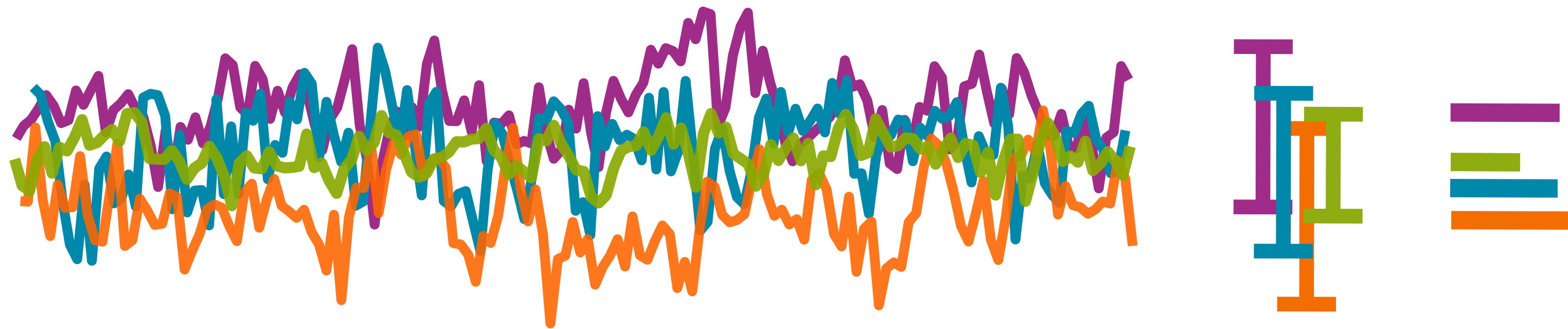


Strong statistical regularities

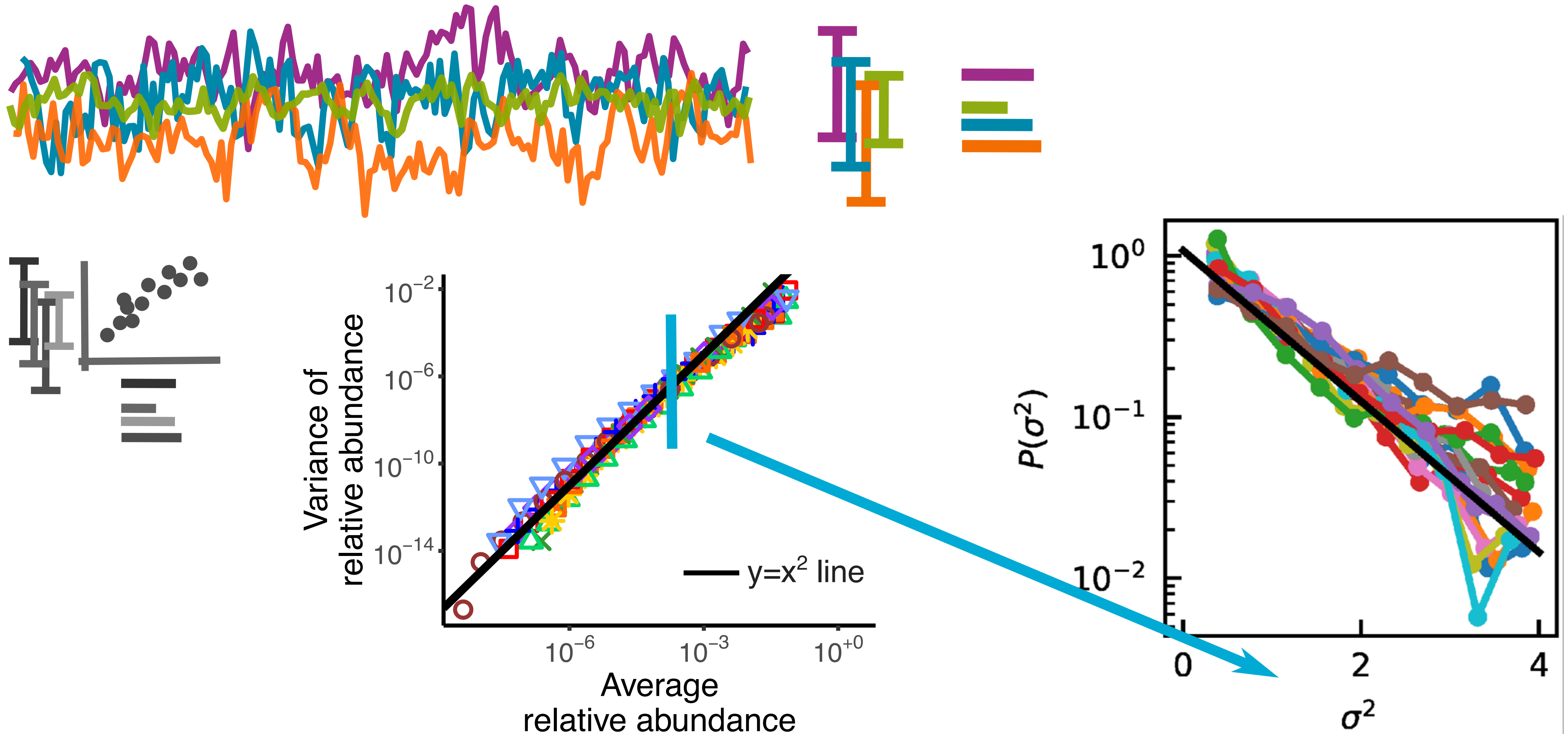




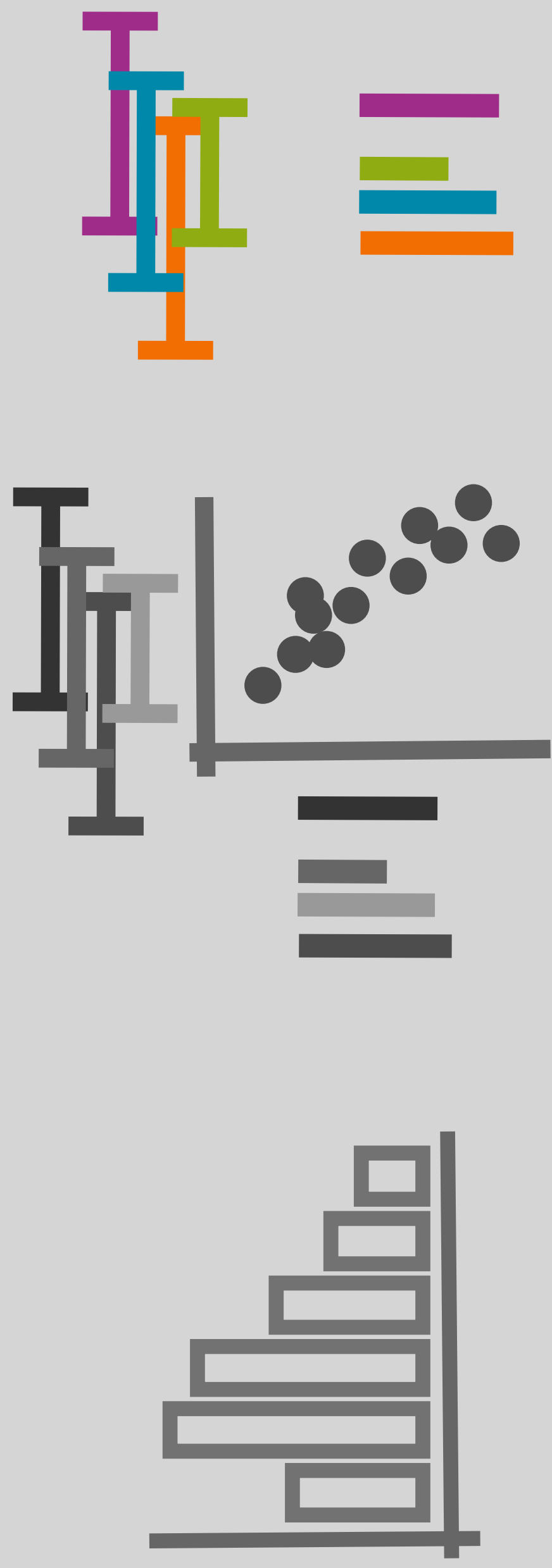
Strong statistical regularities



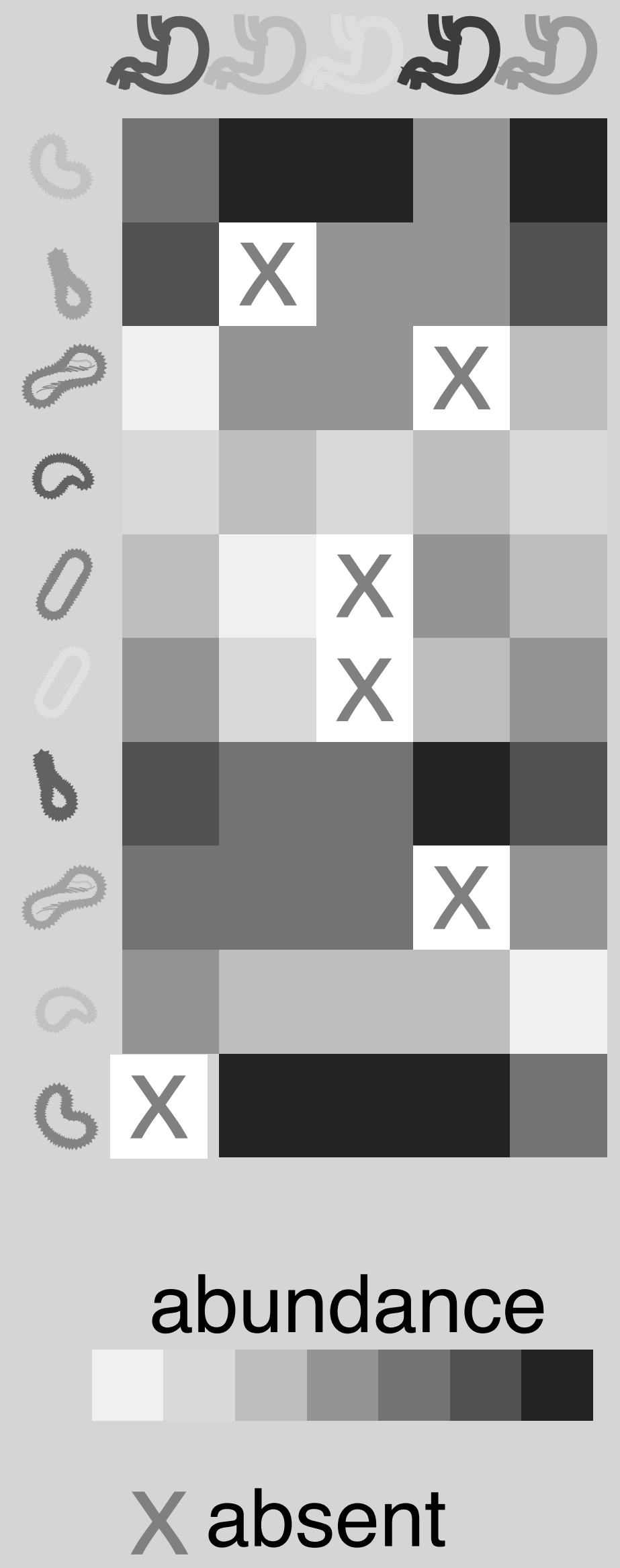
Strong statistical regularities



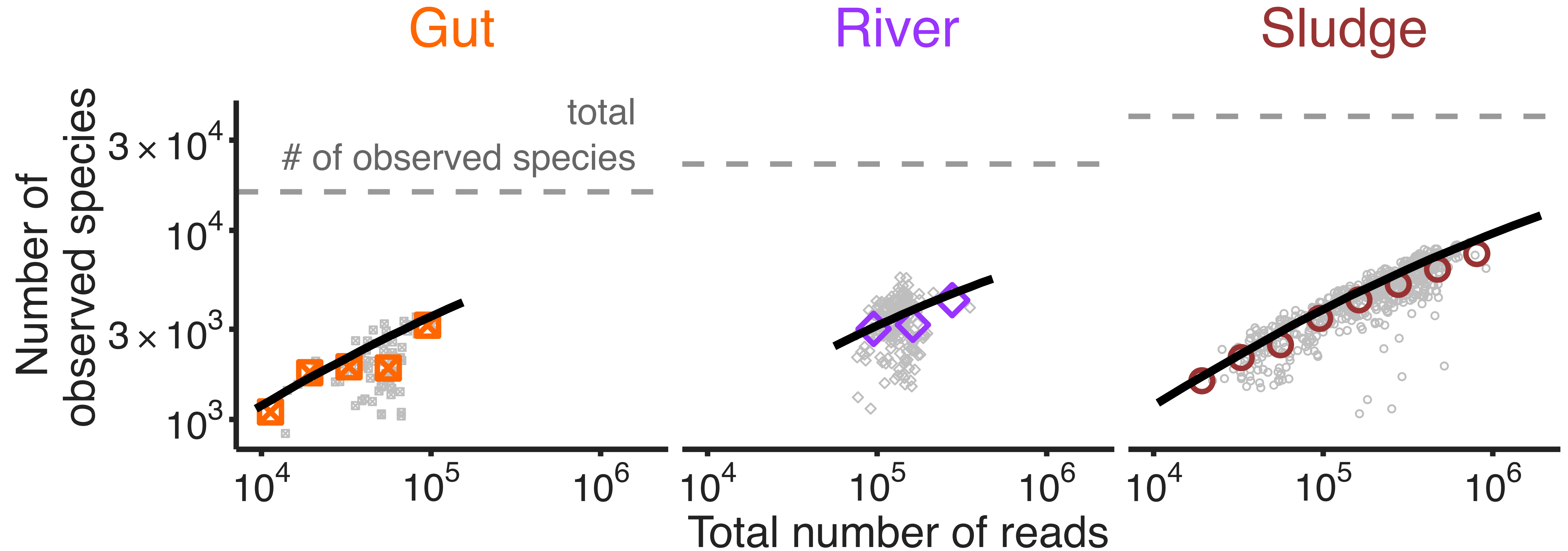
[Grilli, Nat Comm 2020, Zaoli and Grilli Sci Adv 2022]



generates



From statistical regularities to other patterns

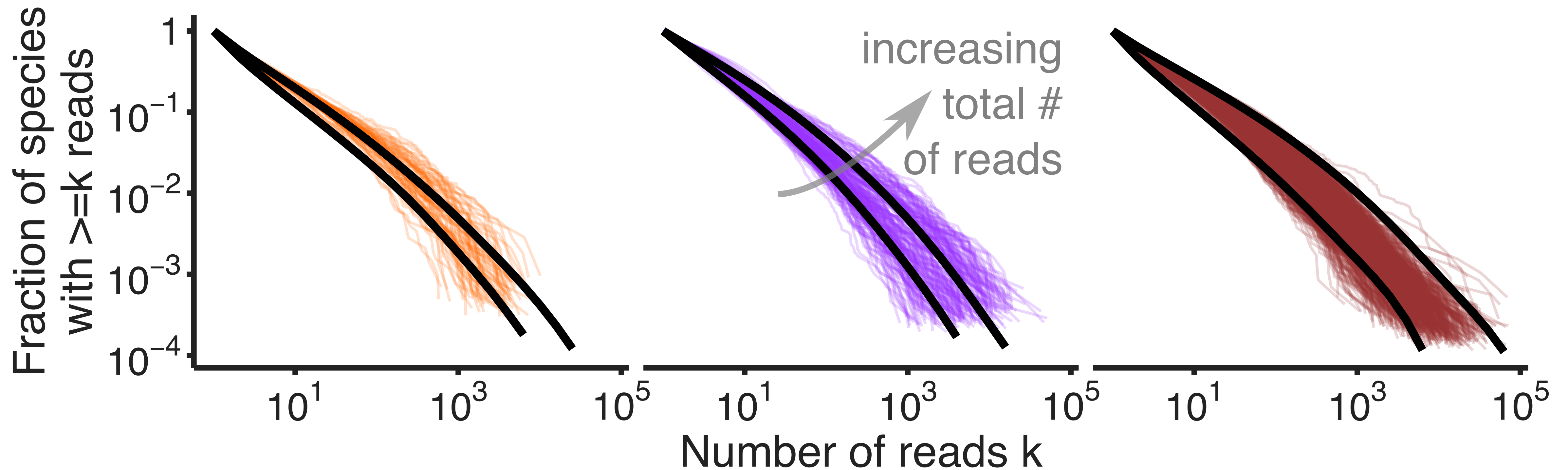


From statistical regularities to other patterns

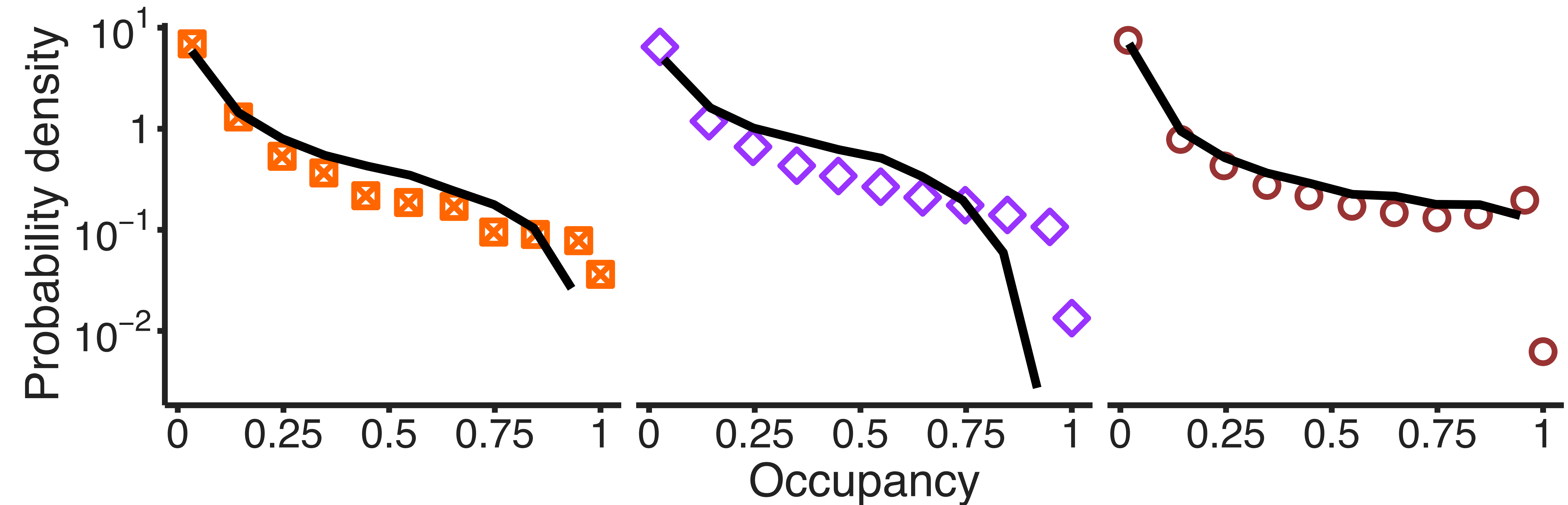
Gut

River

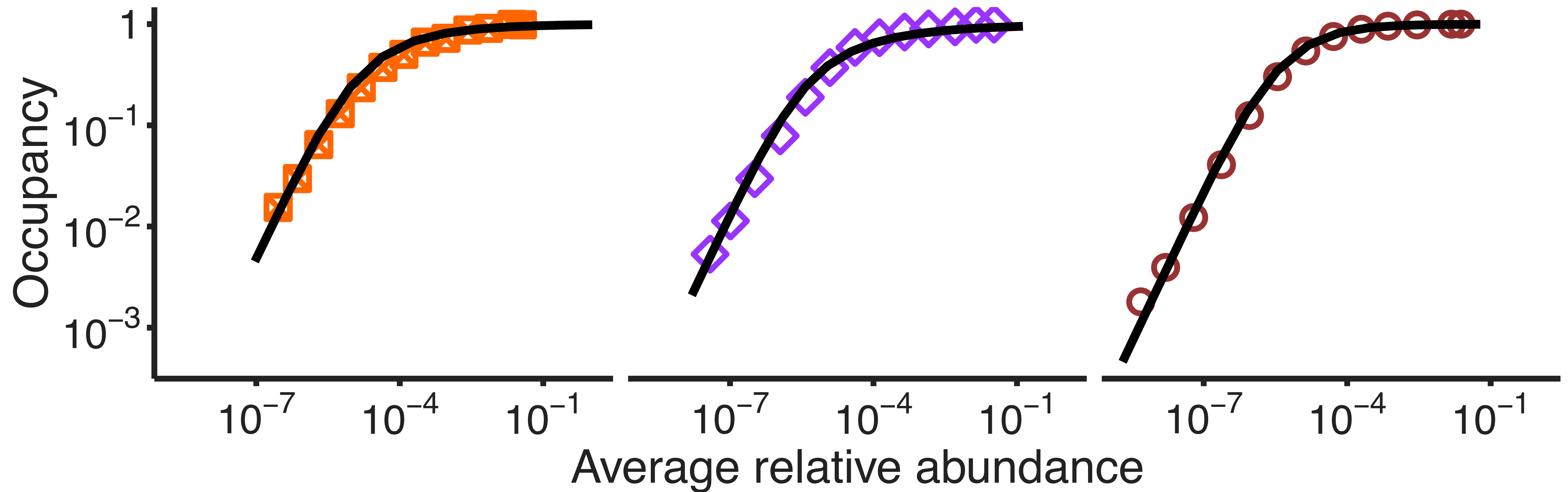
Sludge

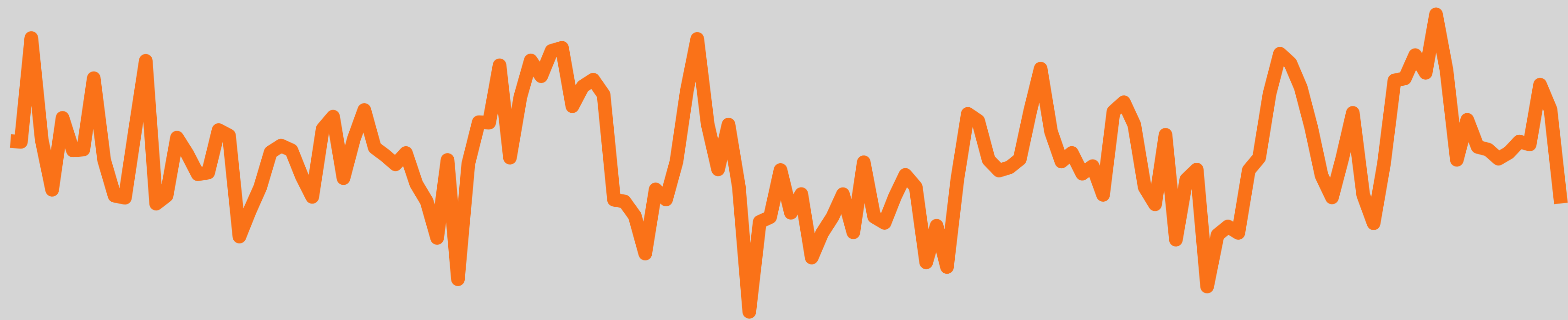


From statistical regularities to other patterns



From statistical regularities to other patterns





Dynamics?

$$\dot{x} = \overset{\text{deterministic}}{\underset{\text{forces}}{f(x)}} + \overset{\text{stochastic}}{\underset{\text{forces}}{\sqrt{D(x)}\xi(t)}}$$

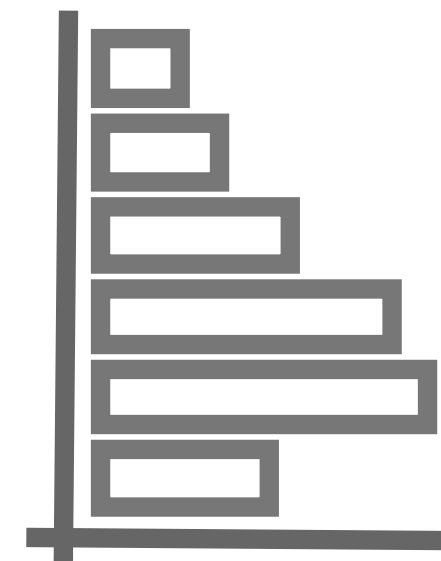
Dynamics?

deterministic
forces

$$\dot{x} = f(x) + \sqrt{D(x)} \xi(t)$$

stochastic
forces

at stationarity



~ gamma

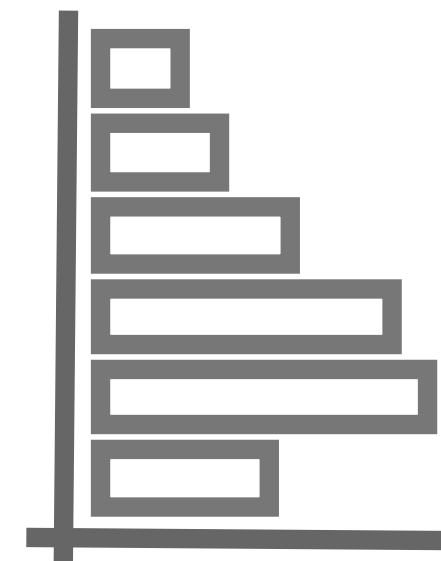
Dynamics?

deterministic
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stochastic
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$$\dot{x} = f(x) + \sqrt{D(x)}\xi(t)$$

at stationarity

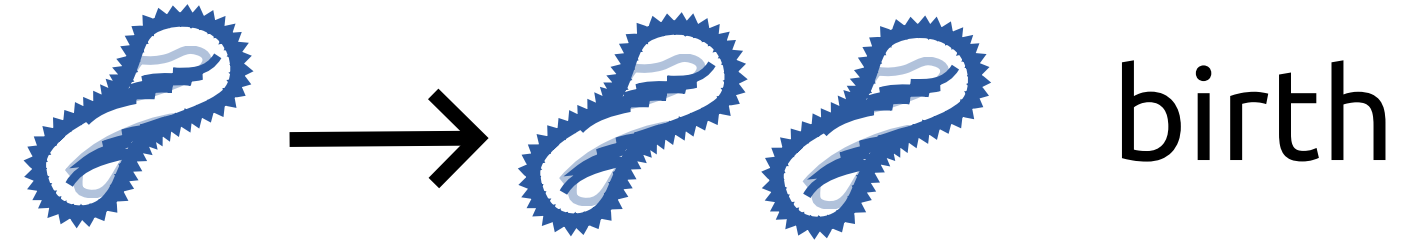


~ gamma

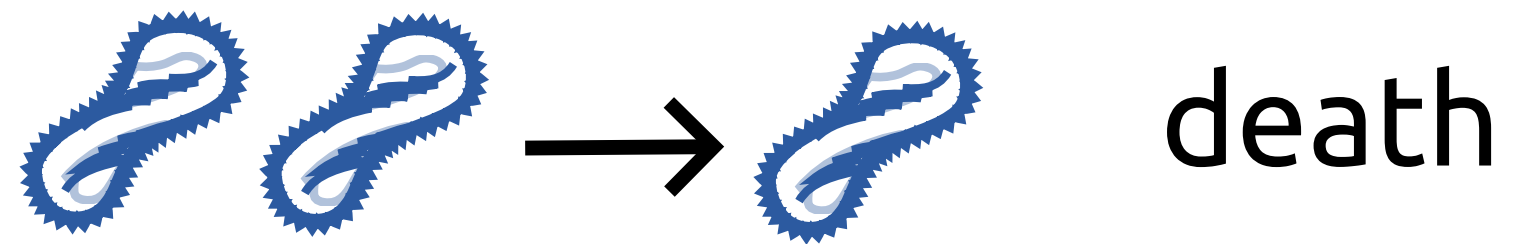
$$-\log D(x) + \int^x dz \frac{f(z)}{D(z)} = a - bx + c \log(x)$$

Two alternative models

Two alternative models



birth

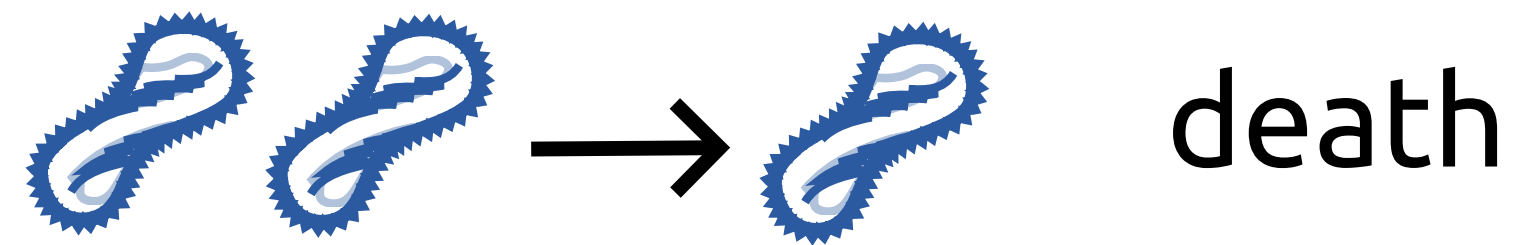


death



migration

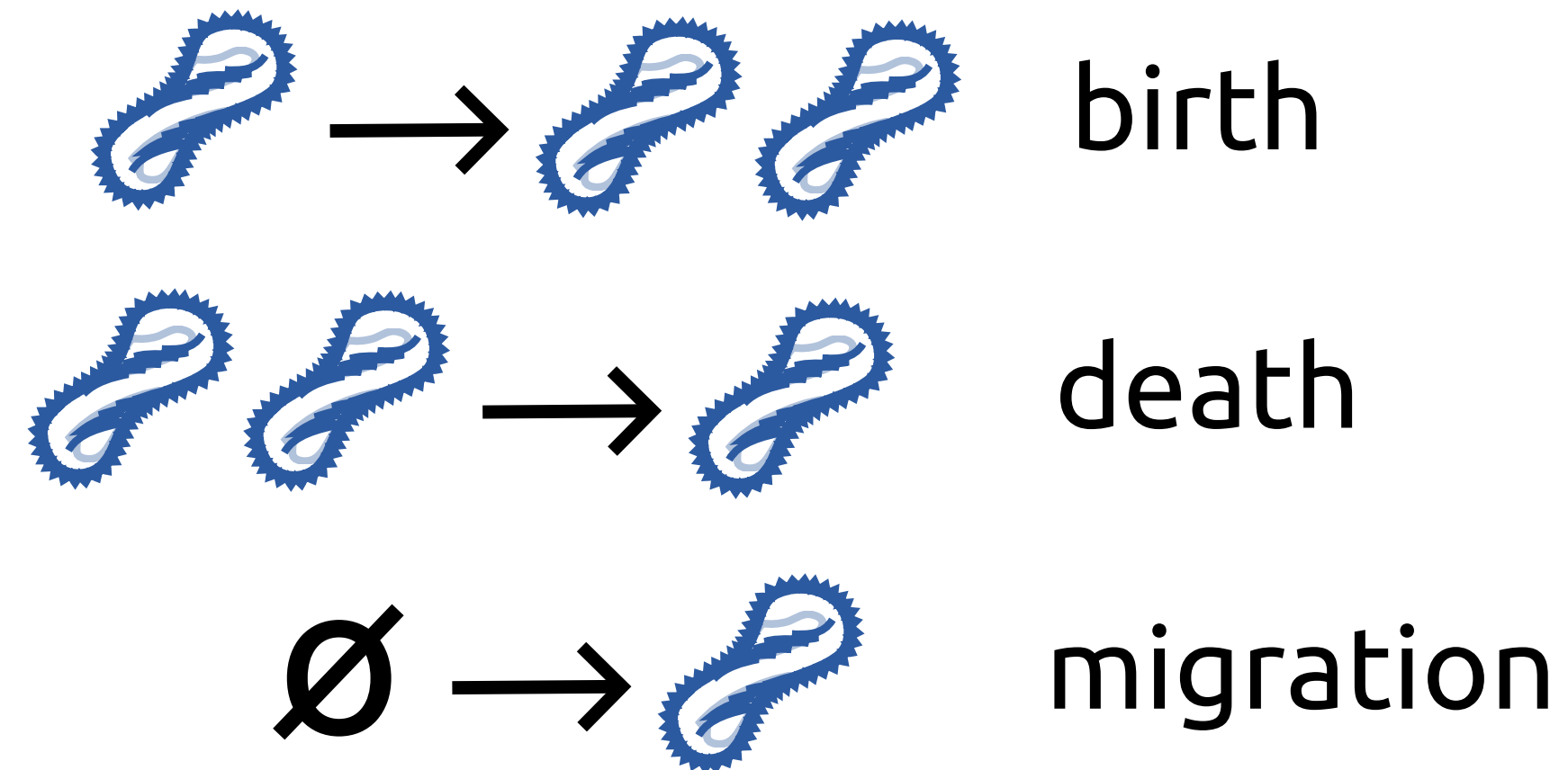
Two alternative models



demographic
stochasticity

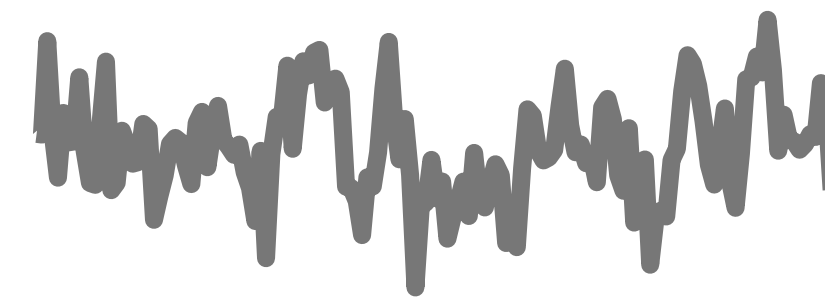
$$\dot{x} = \frac{1}{\tau} (x^* - x) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$

Two alternative models

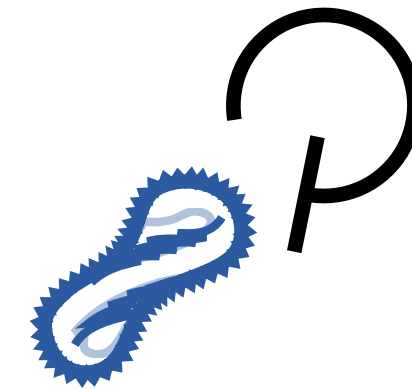


demographic
stochasticity

$$\dot{x} = \frac{1}{\tau} (x^* - x) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$

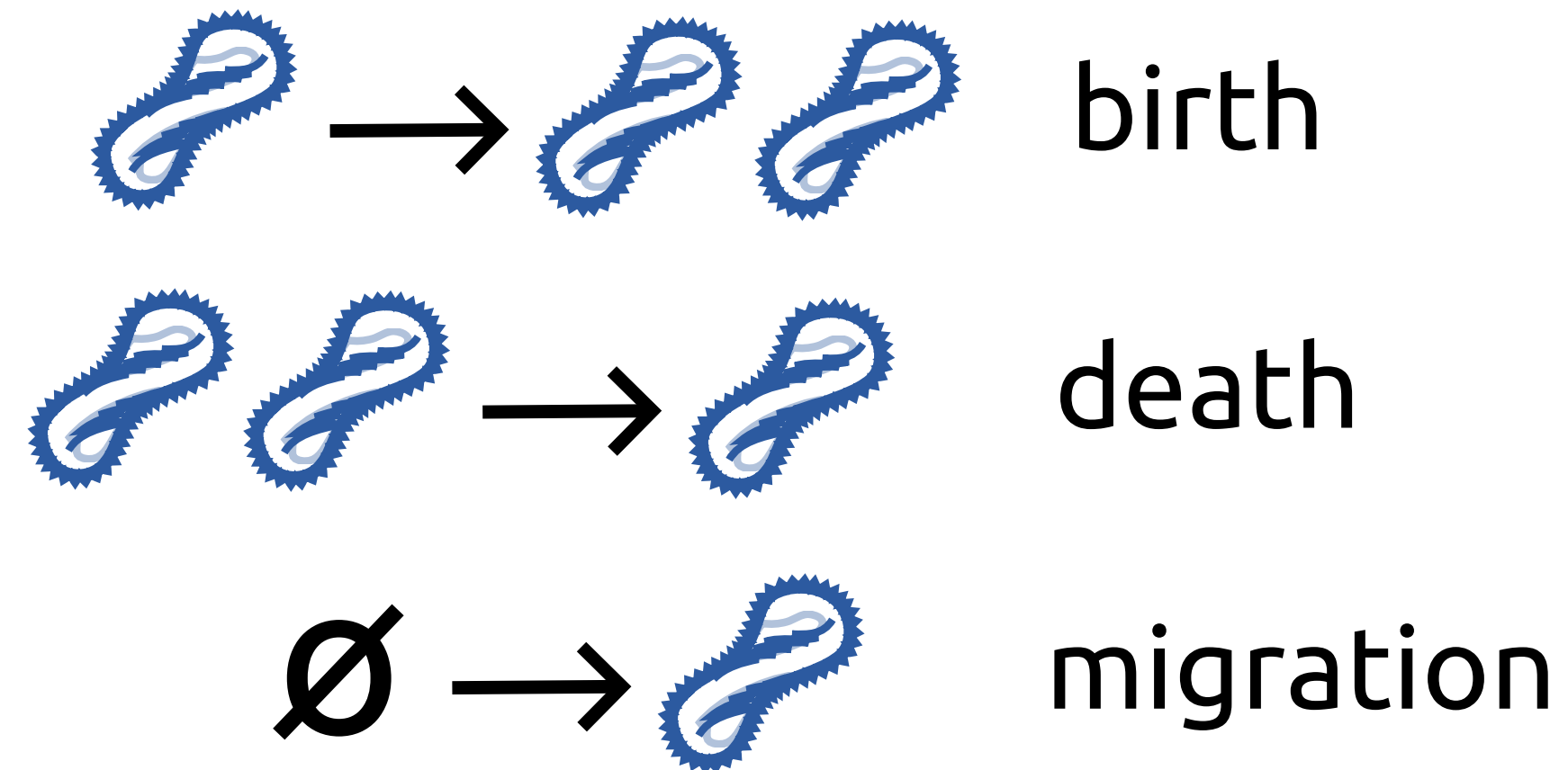


fluctuating
environment



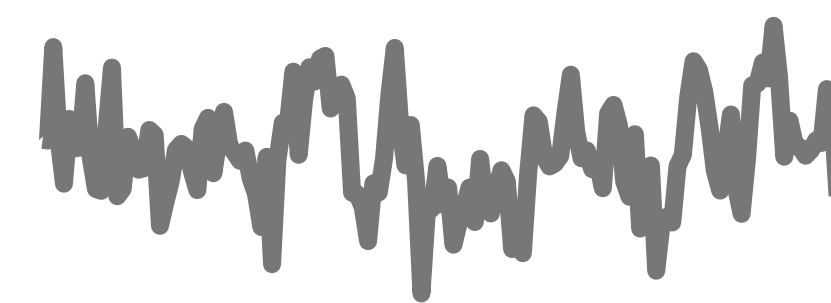
self-limited
growth

Two alternative models

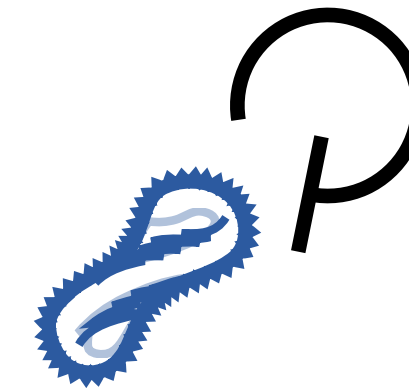


demographic
stochasticity

$$\dot{x} = \frac{1}{\tau} (x^* - x) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$



fluctuating
environment

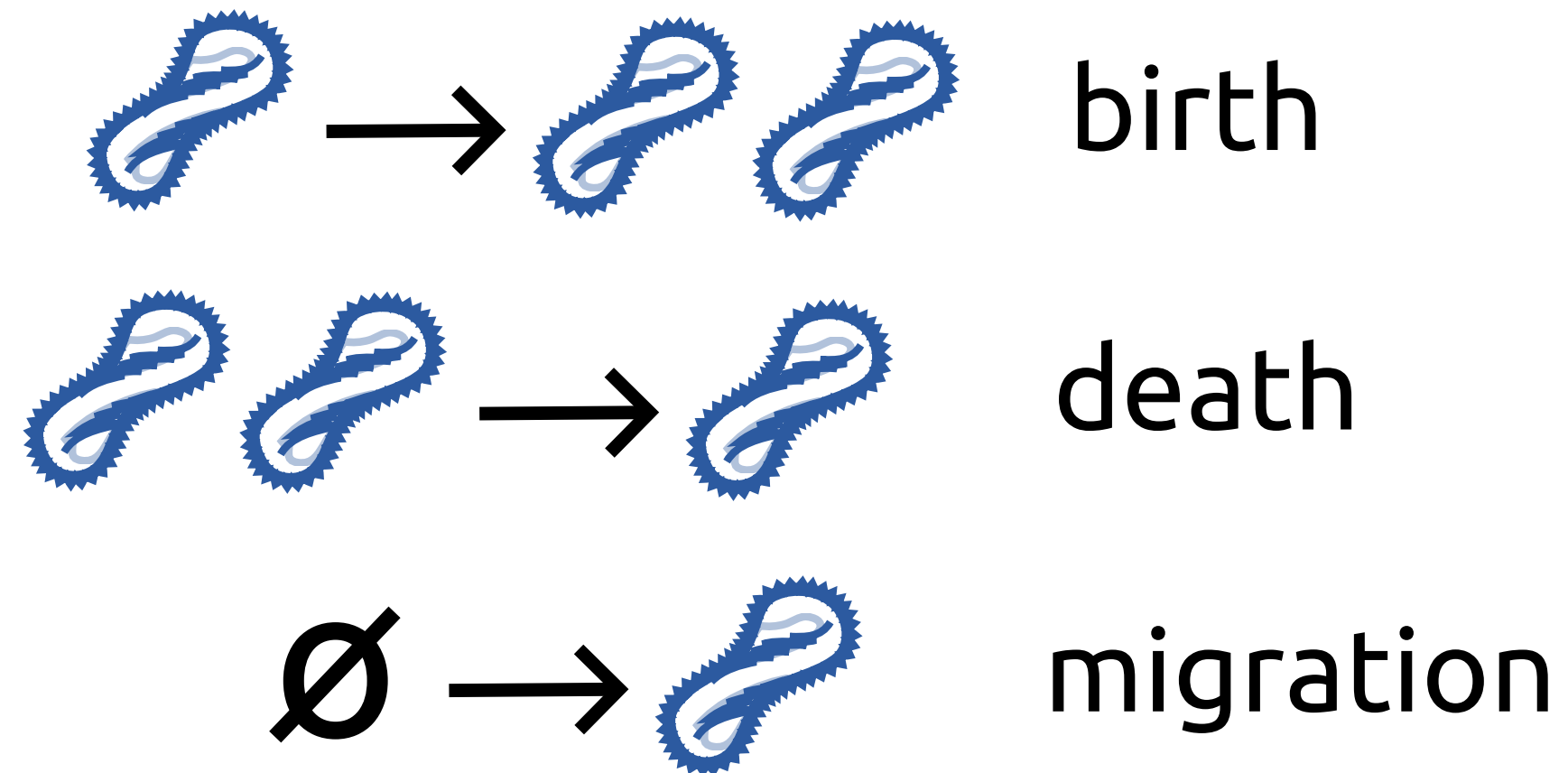


self-limited
growth

environmental
stochasticity

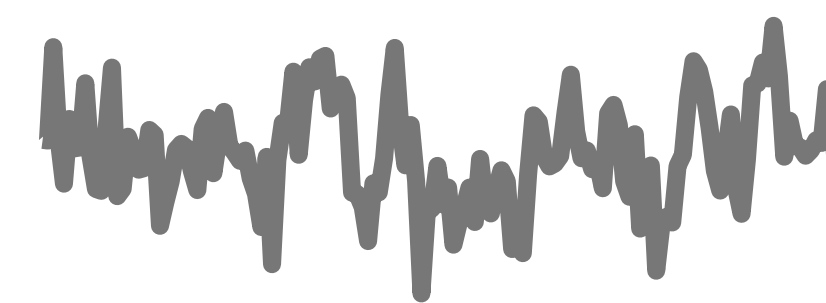
$$\dot{x} = \frac{x}{\tau} \left(1 - \frac{x}{K} \right) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$

Two alternative models

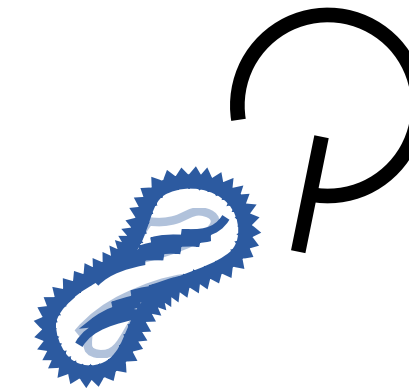


demographic
stochasticity

$$\dot{x} = \frac{1}{\tau} (x^* - x) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$



fluctuating
environment

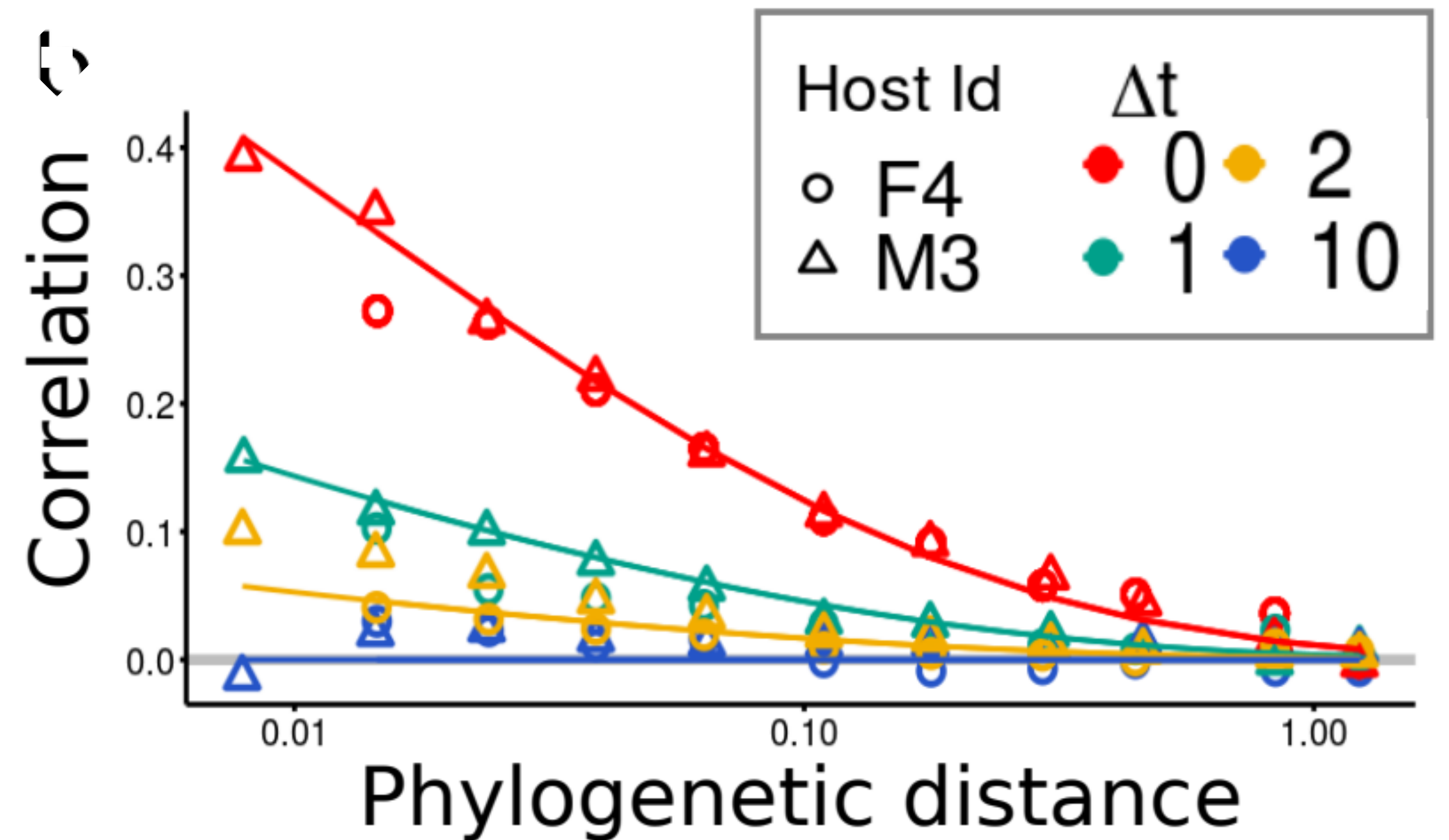
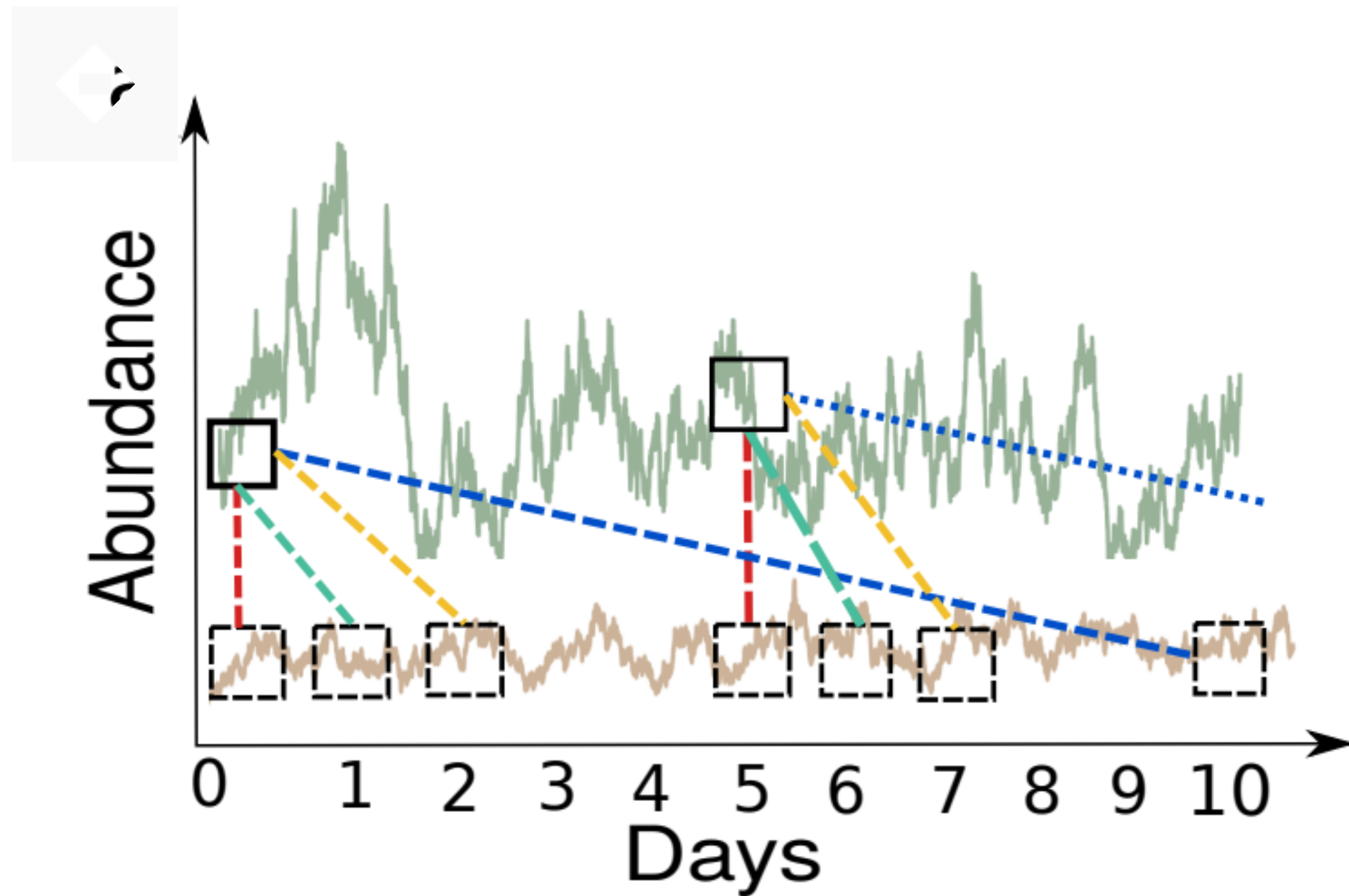


self-limited
growth

environmental
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$$\dot{x} = \frac{x}{\tau} \left(1 - \frac{x}{K} \right) + \sqrt{\frac{\sigma}{\tau}} x \xi(t)$$

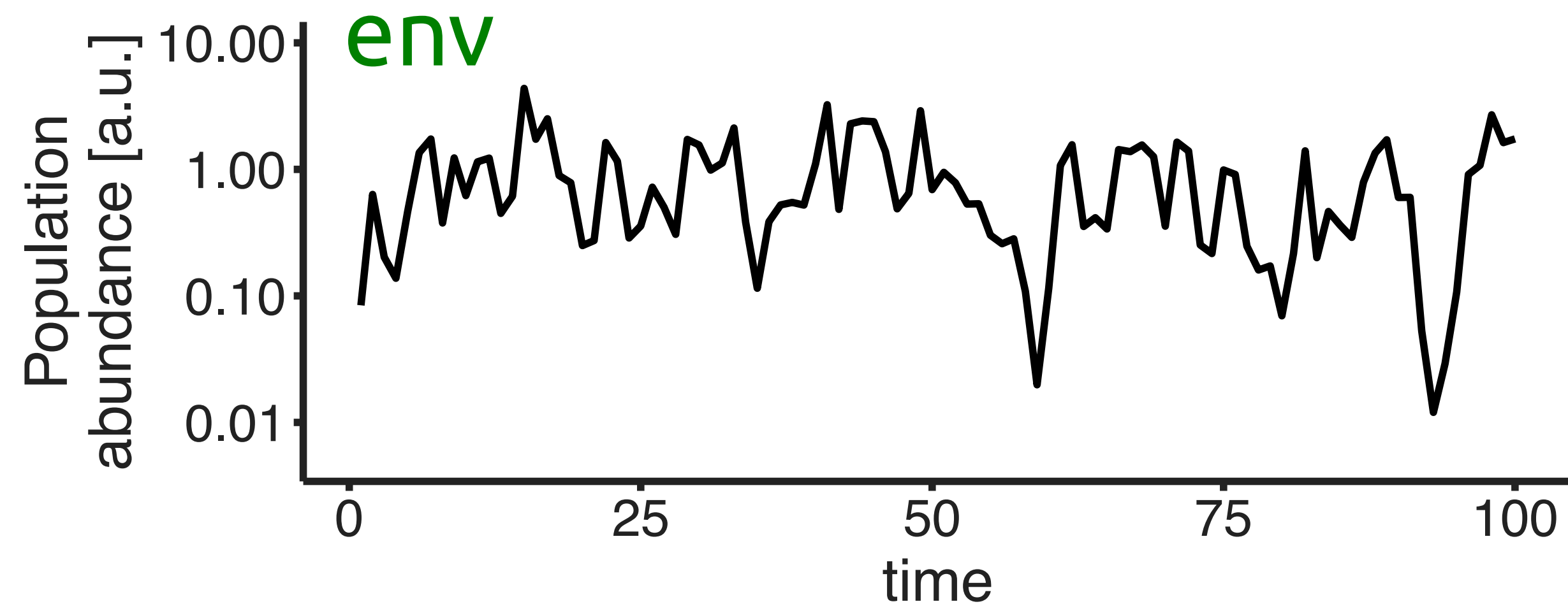
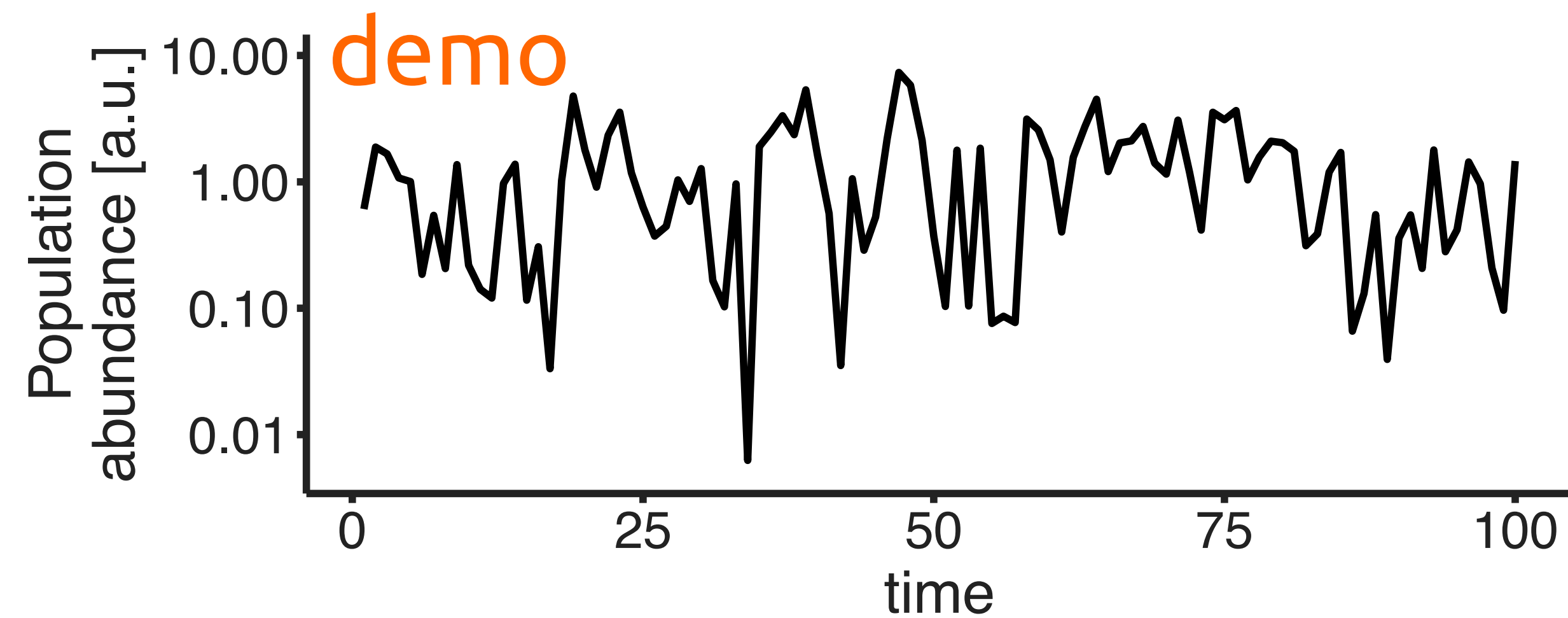
Similar species are positively correlated consistently with environmental fluctuations



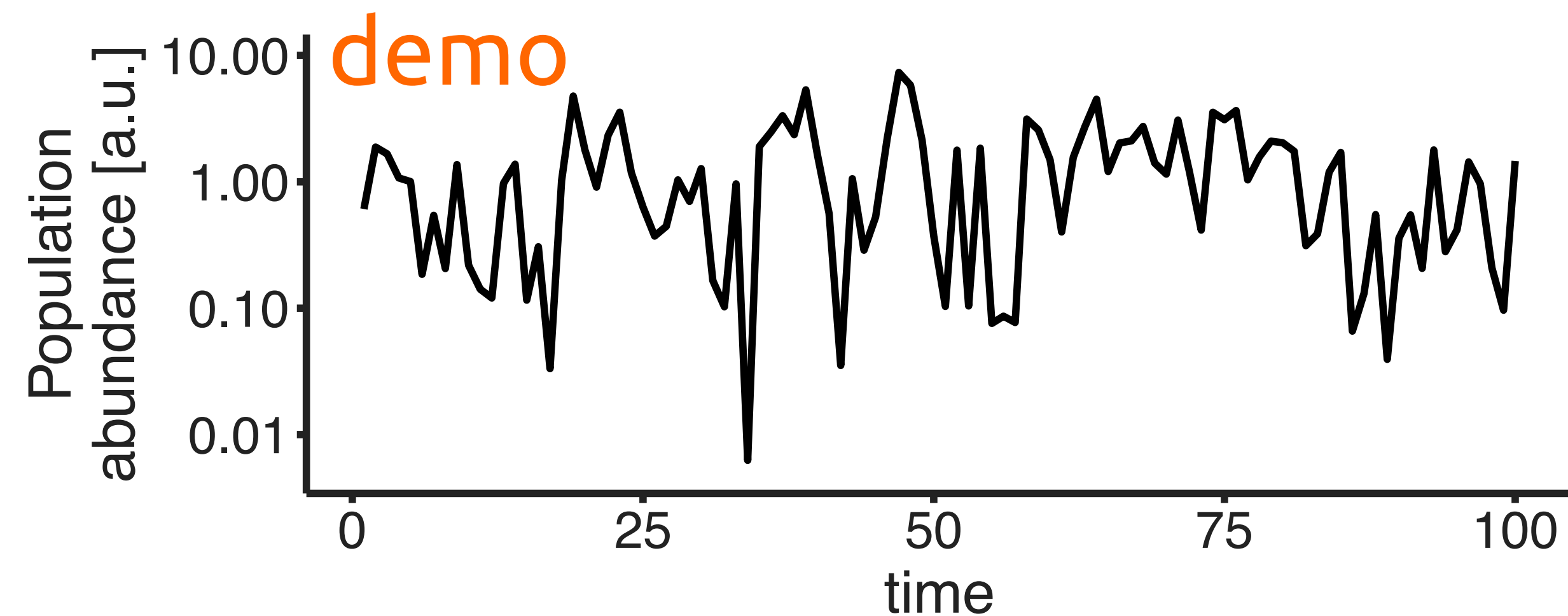
Sireci, Muñoz, Grilli PNAS 2023

**...indirect evidence
what about dynamics?**

How do we differentiate between trajectories



How do we differentiate between trajectories

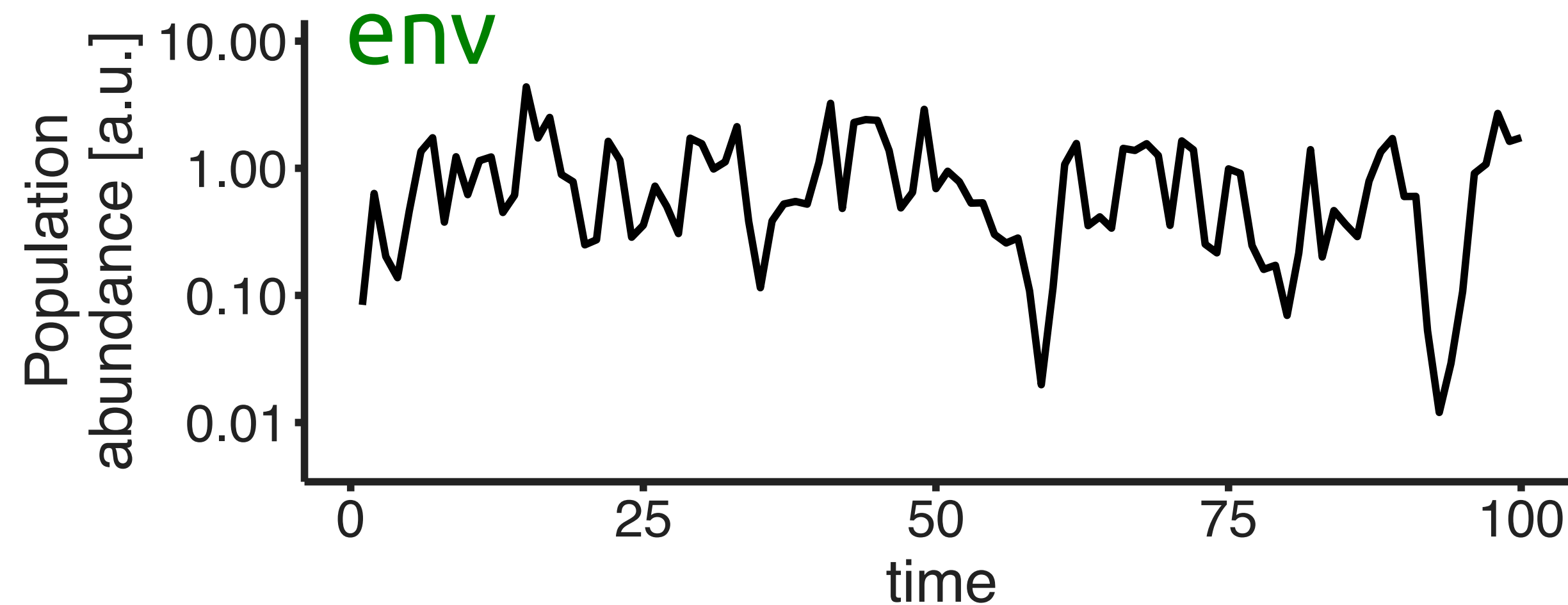


different

$$P(x, t + \tau | x_0, t)$$

same

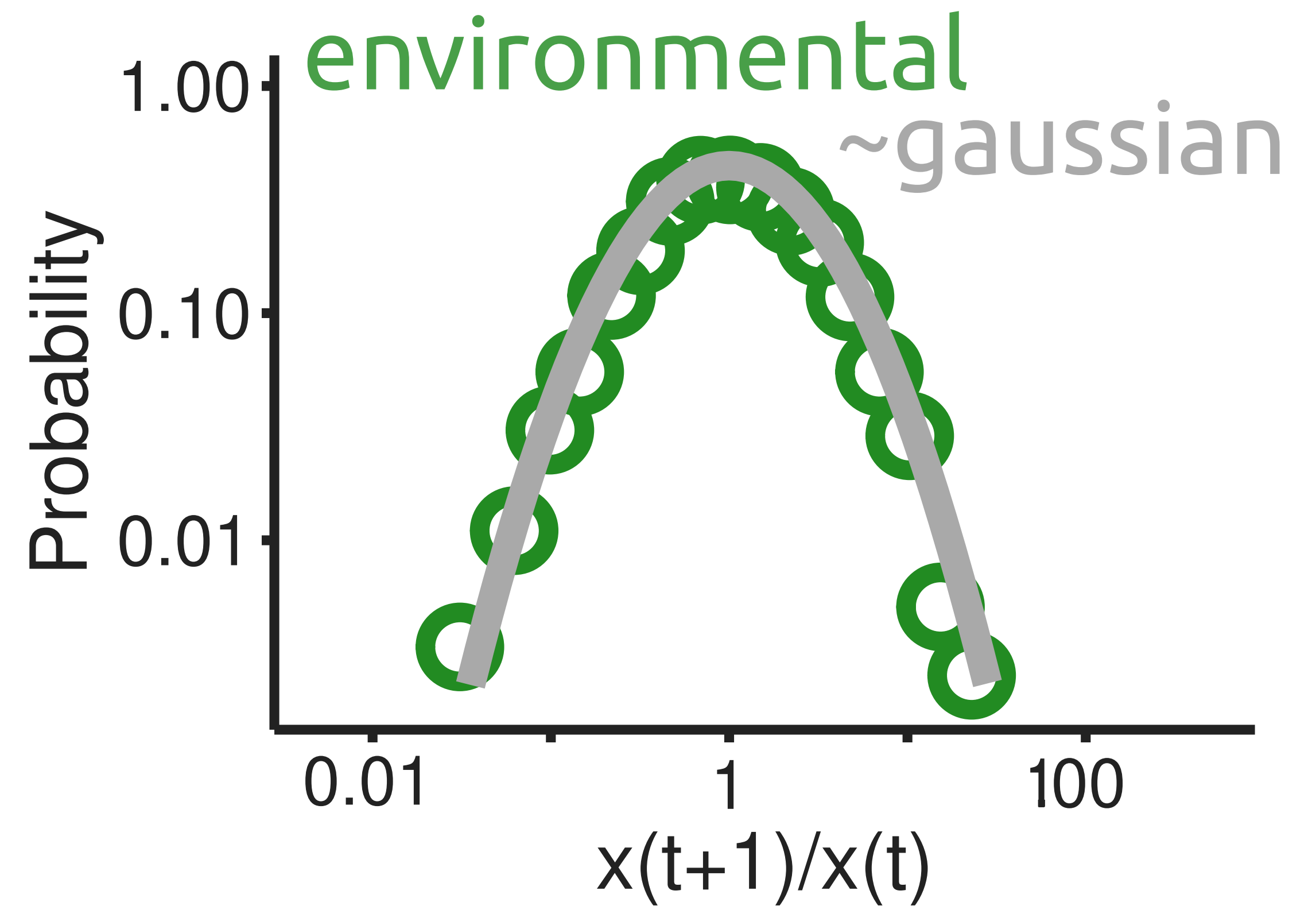
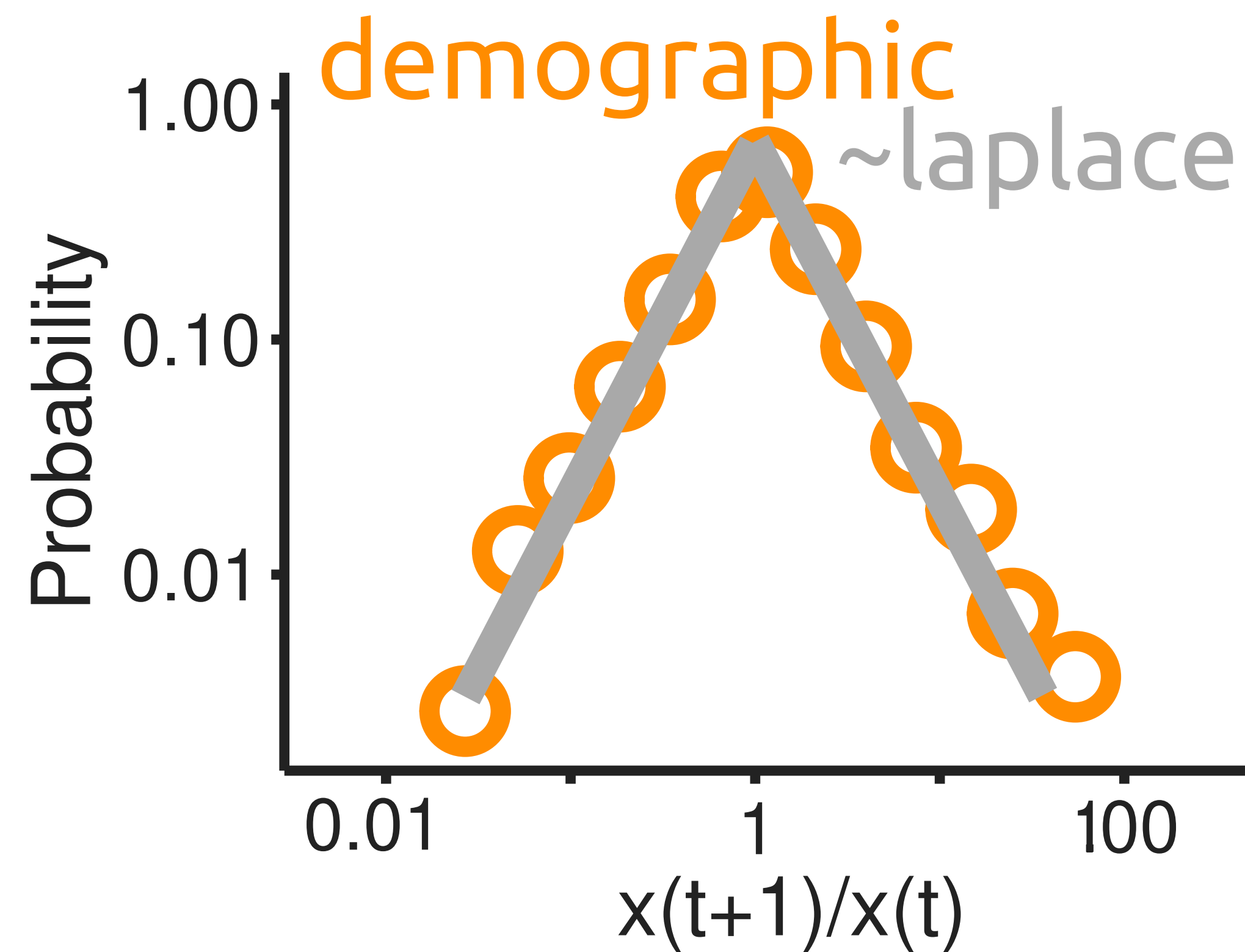
$$P^*(x) = \lim_{t \rightarrow \infty} P(x, t | x_0, 0)$$



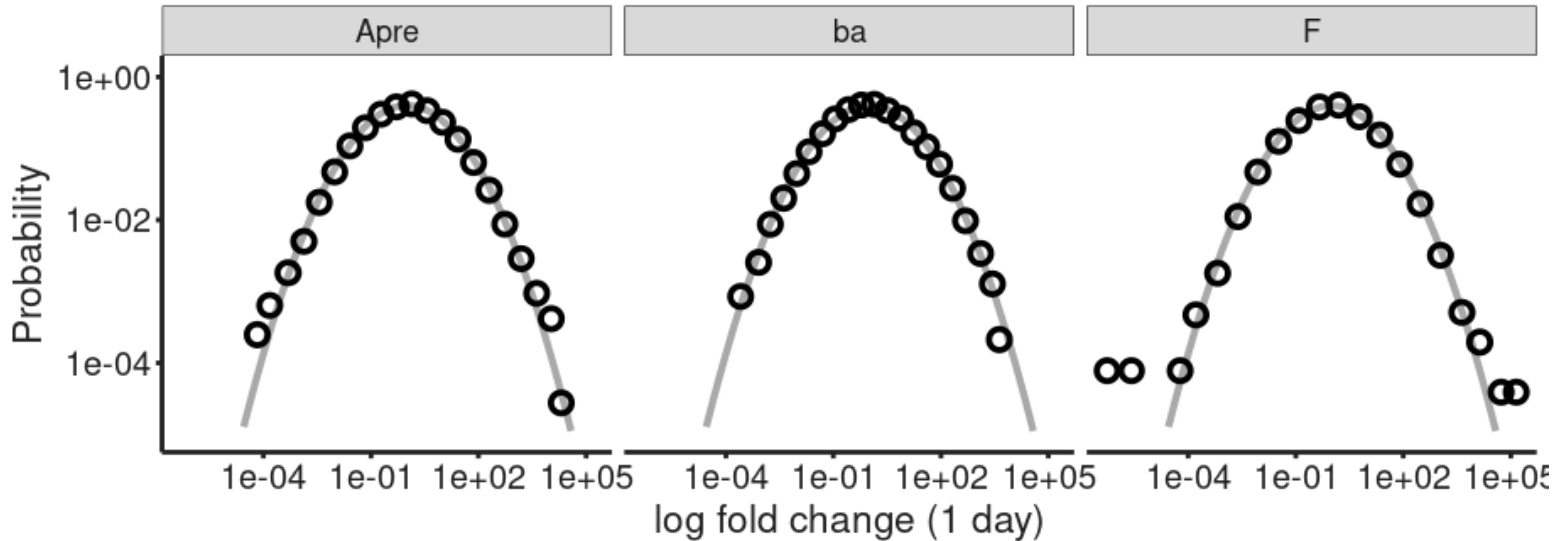
Alternative predictions for log change distr

$$r = \log \left(x(t + \tau) / x(t) \right)$$

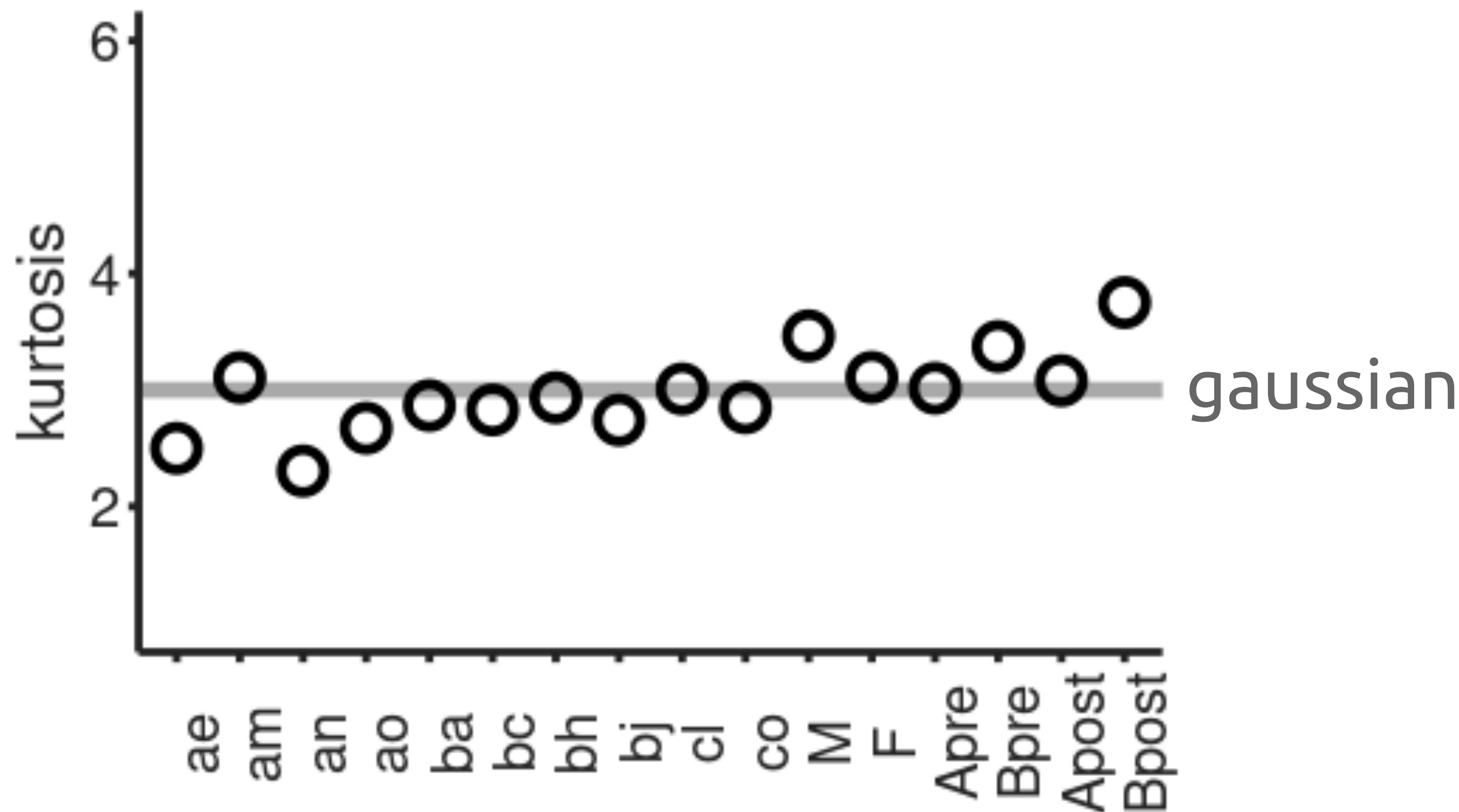
$$P^*(r) = ?$$



Log-rate change is Gaussian in data



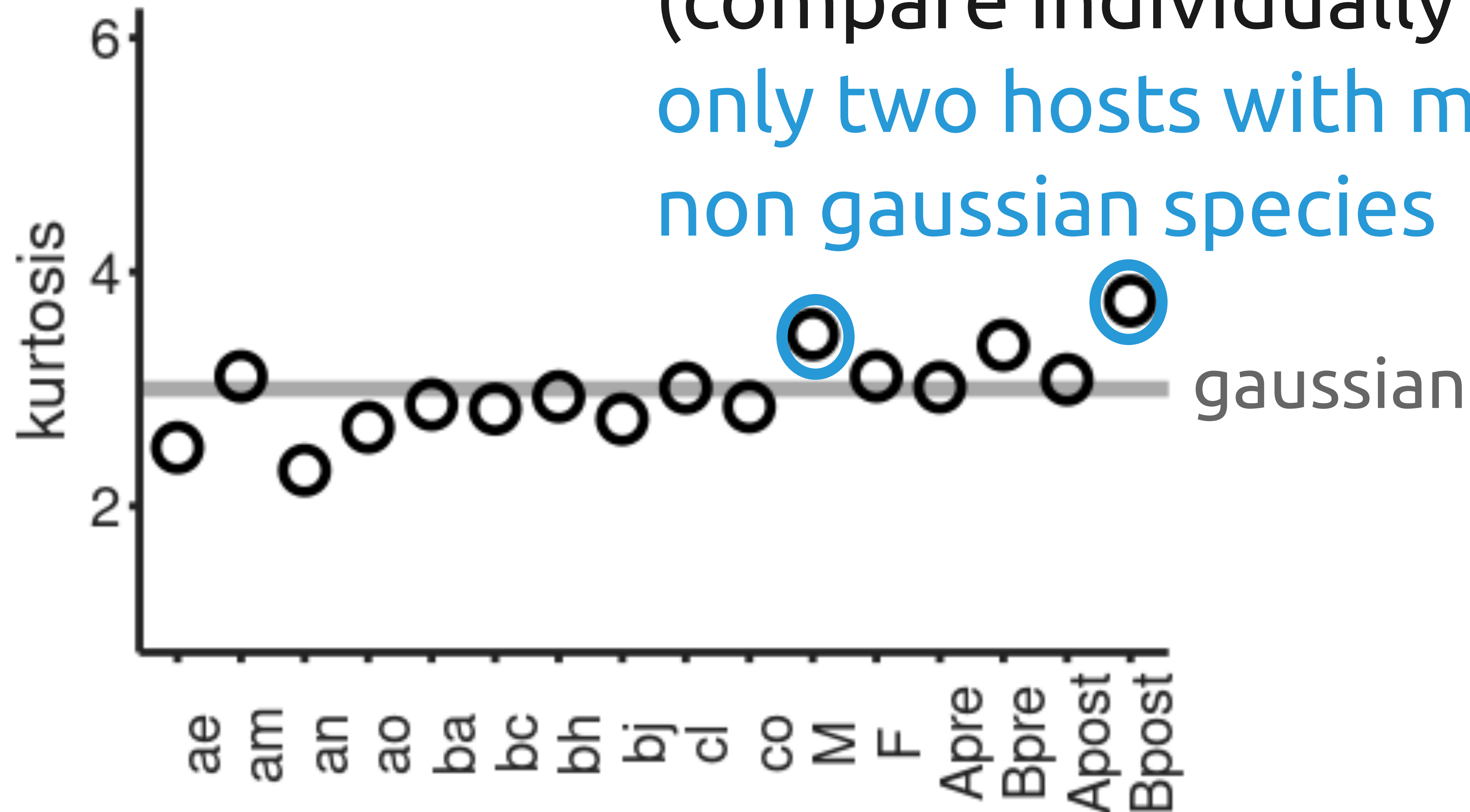
Log-rate change is Gaussian in data



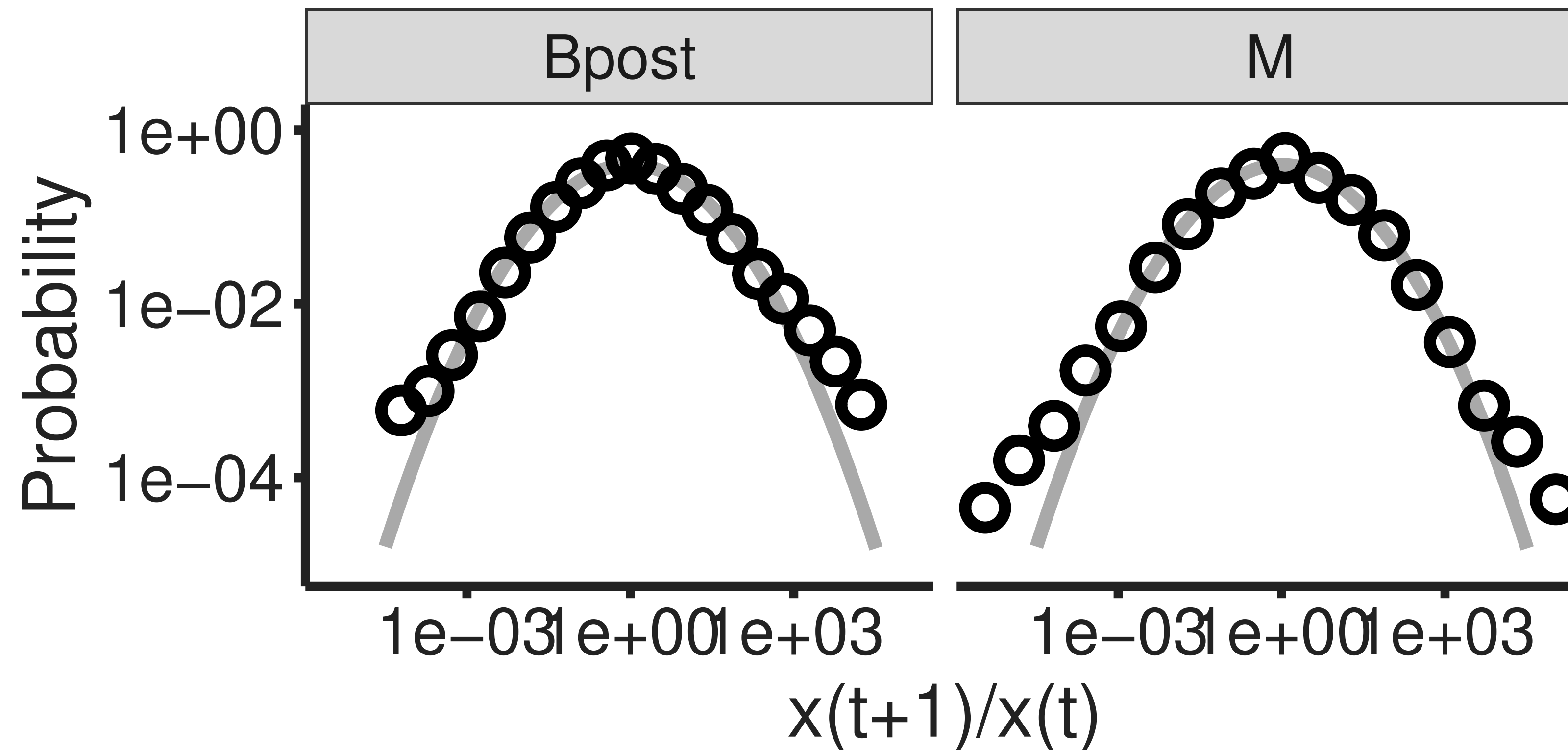
Log-rate change is Gaussian in data

(compare individually per sp)

only two hosts with more
non gaussian species

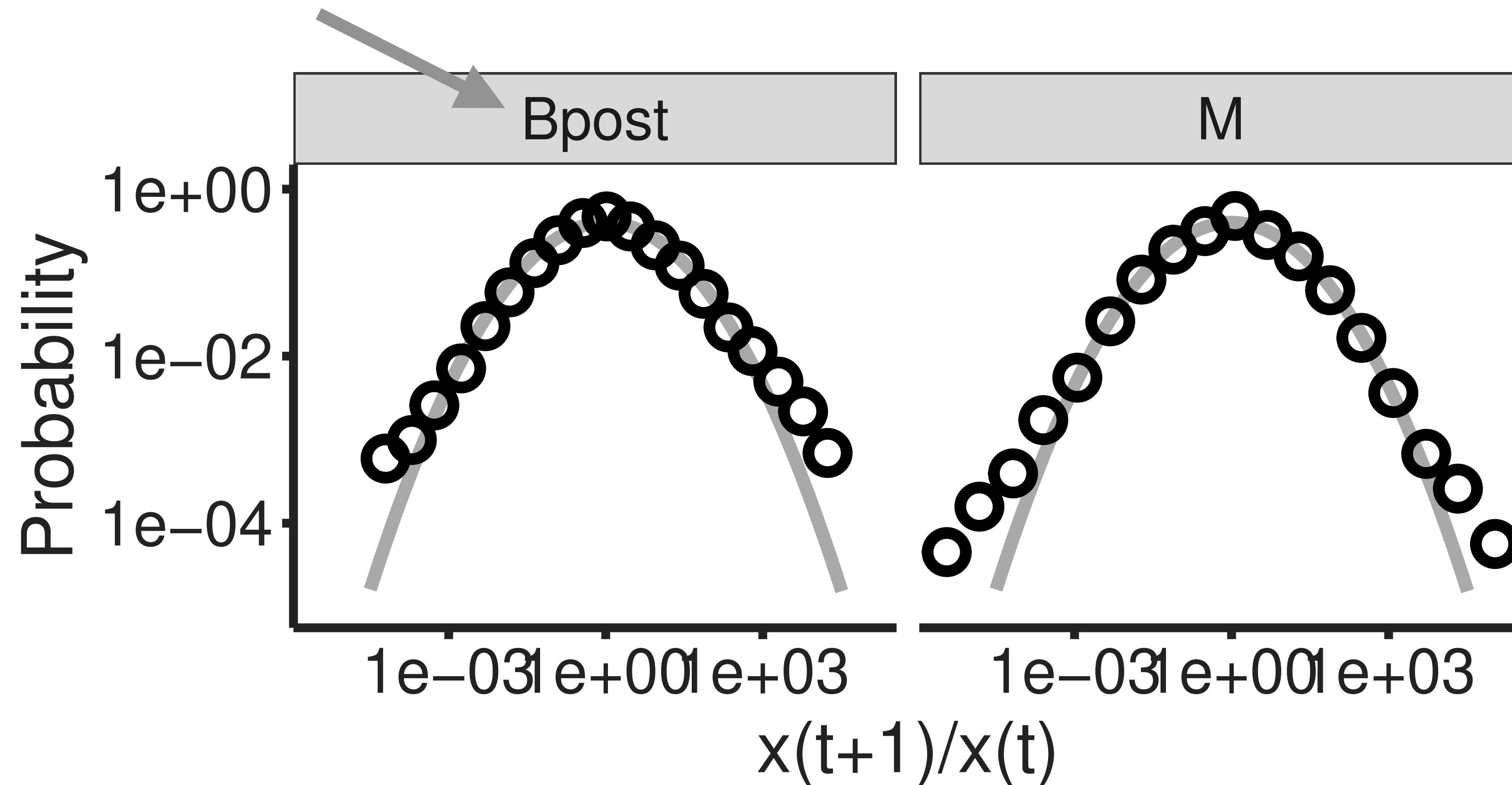


Two hosts have non-gaussian log-fold changes

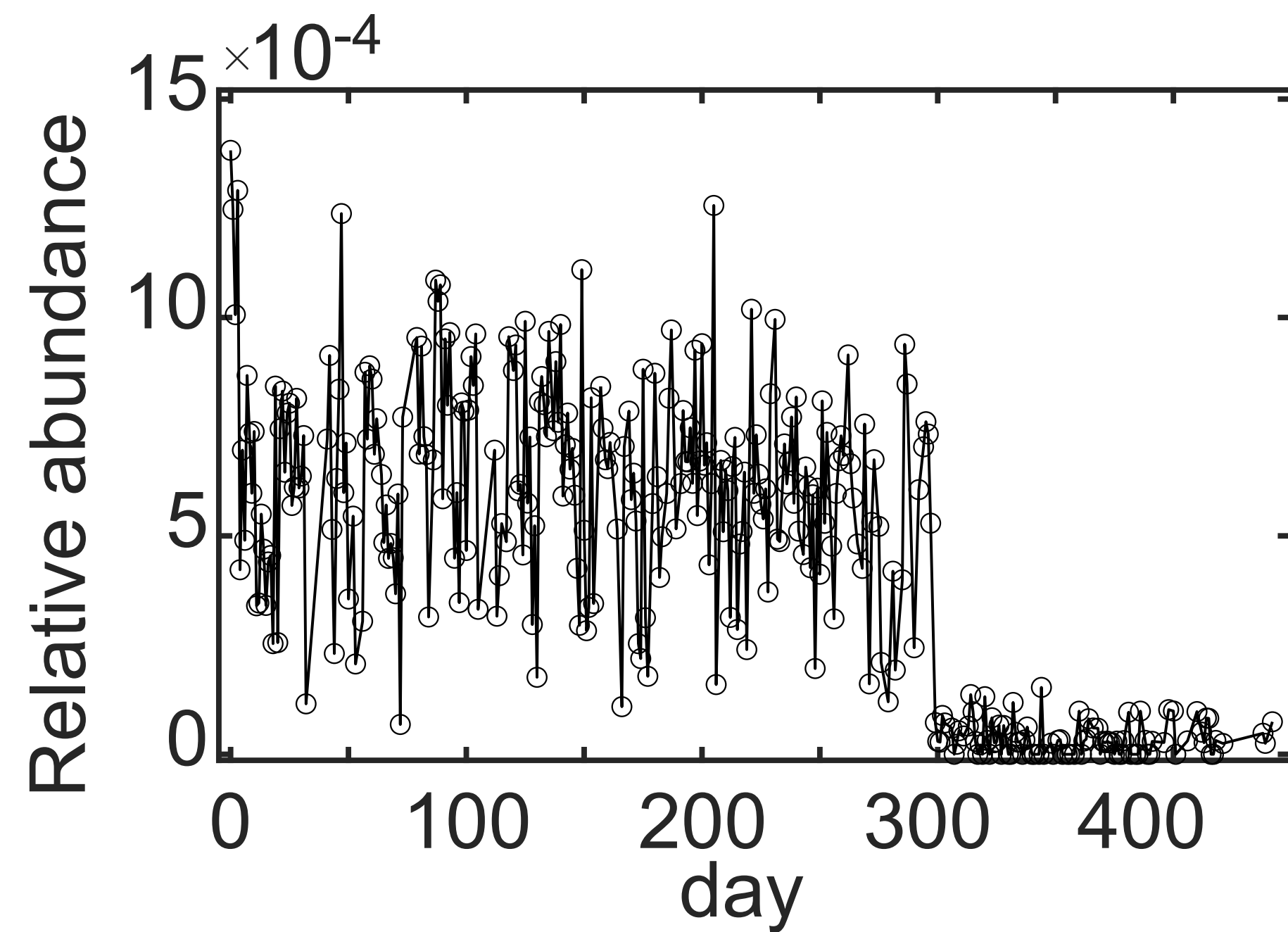


Two hosts have non-gaussian log-fold changes

post salmonella infection

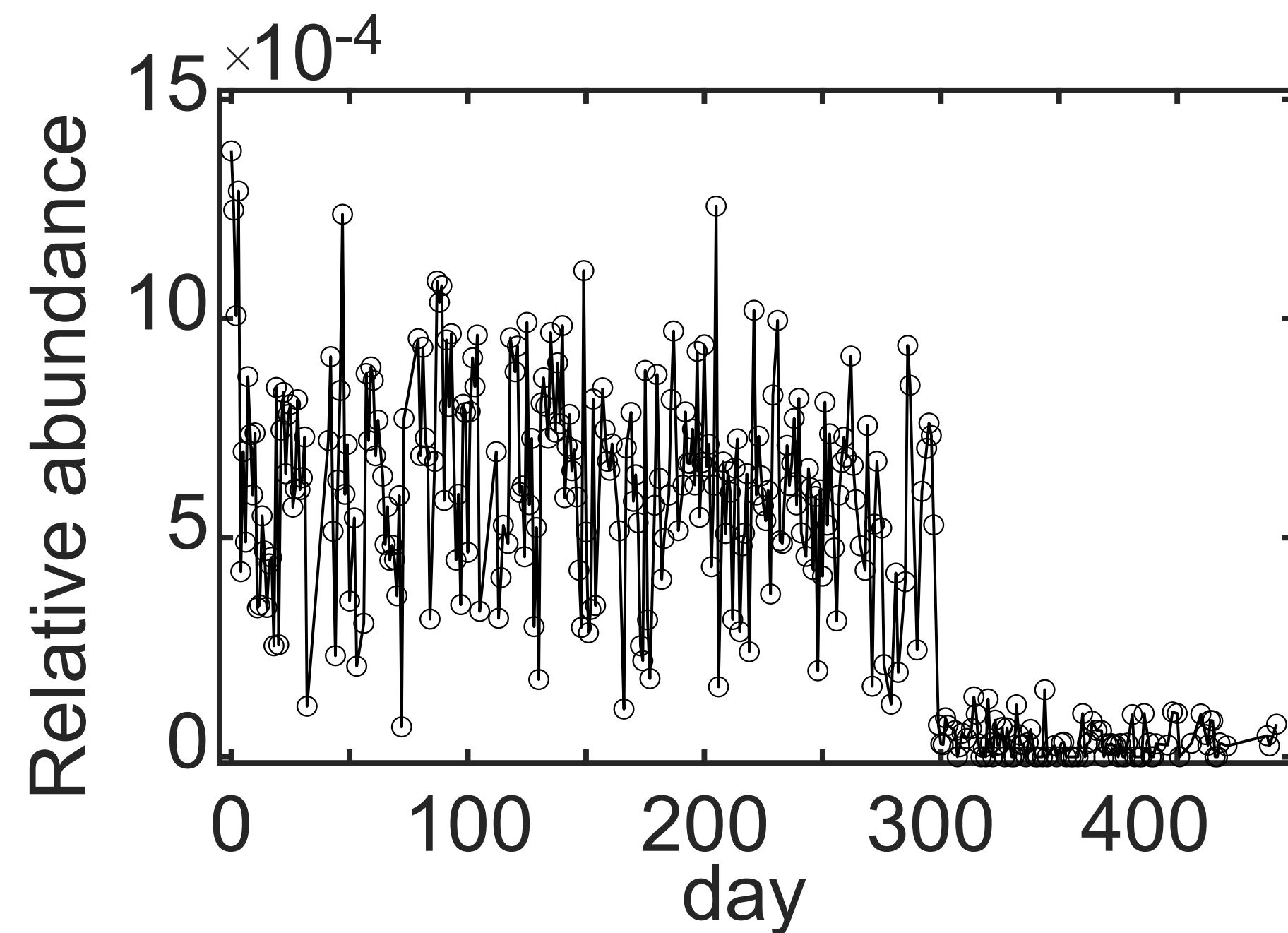


There exist unstable species on long timescales

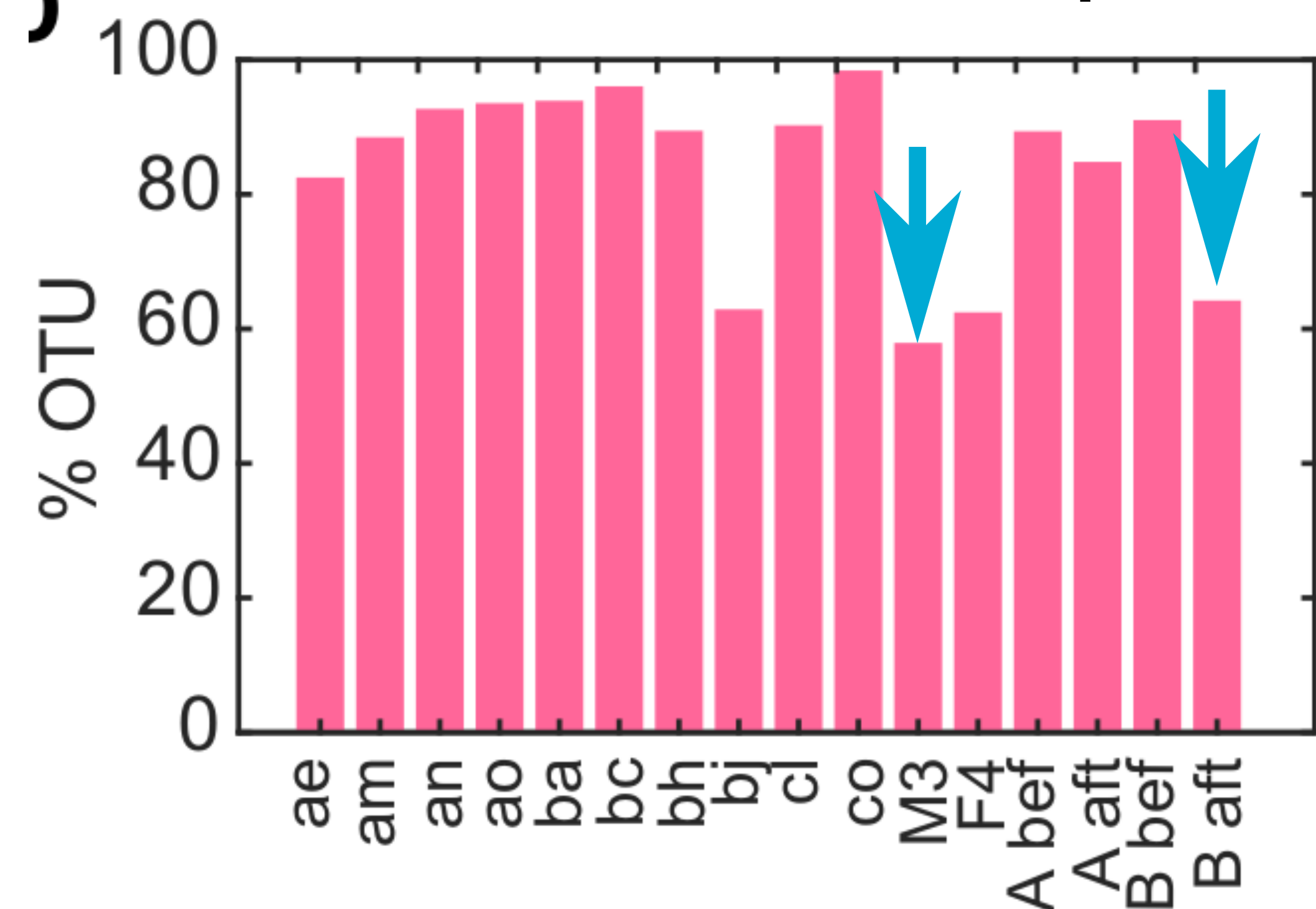


[Zaoli & Grilli, Science Advances 2021]

There exist unstable species on long timescales



fraction of "stable" species



Removing unstable species recover gaussianity

